Do Sparse Autoencoders Preserve What a Language Model Uses?

A Layer-wise Audit of Reconstruction, Information, Behavioural and Causal Faithfulness in GPT-2 Small

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Abstract

Sparse autoencoders (SAEs) are widely used to explain what large language models represent, which presumes that an SAE is a faithful stand-in for the activations it replaces. We audit that presumption at all twelve layers of GPT-2 Small for three SAE architectures (ReLU, TopK and a sigmoid-surrogate JumpReLU; 108 SAEs at matched sparsity, three seeds each), separating four levels of faithfulness: reconstruction, task-information preservation, behavioural faithfulness and causal faithfulness. Reconstruction degrades with depth (OWT test explained variance 0.93–0.95 at layers 0–2, 0.76–0.82 at layer 11). Within a layer, reconstruction error predicts how much splicing the SAE damages the model's predictions, but not how much truth information is lost. The apparent collapse of a frozen truth probe on SAE reconstructions is largely reproduced by random noise of the same norm, but for persona instructions the SAEs lose 2–3 times more than such noise and re-trained readers recover almost none of it; at equal reconstruction error the SAEs never preserve task information better than PCA, and they distort the honest-minus-dishonest activation difference (per-pair cosine 0.35–0.58). Using an equal-length persona-instruction task built to remove a length shortcut, we locate where GPT-2 carries the instruction's effect and decompose it: the SAE-reconstructed activation difference restores 93–97 % of the effect at the layer-1 persona token but only 62–68 % at the layer-8 and layer-11 final token, with the remainder in the reconstruction error or in an interaction between reconstruction and error. Short fine-tuning at layers 3, 6, 8 and 9 separates these levels again. Reconstruction fine-tuning on truth statements cuts their reconstruction error by 61–85 % and lowers the truth gap measured by a frozen nonlinear probe fitted on original activations, meeting its pre-registered criterion in 4 of 9 layer–architecture cells beyond layer 6; frozen logistic and re-trained probes show no consistent gain, so we do not claim that truth information is recovered, and it leaves persona representation and causal preservation unchanged. Adding an output-KL term improves behavioural faithfulness consistently in all 9 cells, but only 5 are creditable: in the other 4 (ReLU and JumpReLU at layers 8–9) L0 rose 5.3–8.1 %, beyond the pre-registered ±5 % guard. Output-KL never improves truth preservation, and its apparent gains in causal preservation (+0.02 of the effect) occur only in those sparsity-violating cells; for TopK, whose sparsity is fixed, it has no effect. All results except the reconstruction table are VALIDATION findings, not confirmed results; a single pre-registered held-out evaluation is pending.

PENDING: Abstract to be revised after E1, E2 and the final held-out evaluation (M5). Numbers above are validation-stage unless marked TEST.

1 Introduction

Sparse autoencoders decompose a model's dense activations into a larger set of sparsely active latents, many of which look human-interpretable [3, 4]. They have become a standard tool of mechanistic interpretability, with architectures that trade off reconstruction against sparsity in different ways [5, 13]. Interpretability claims made through an SAE — that the model represents truth with certain features, or that a feature controls a behaviour — inherit an assumption: that the SAE is a *faithful* replacement for the activation it reconstructs.

Faithfulness has several distinct meanings that are often conflated:

1. **Reconstruction**: the reconstruction $\hat{x}$ is close to the activation x.
2. **Information preservation**: task information that the model encodes (here, whether a statement is true; whether a persona instruction is honest) can still be read from $\hat{x}$.
3. **Behavioural faithfulness**: the model behaves the same when $\hat{x}$ replaces x.
4. **Causal faithfulness**: $\hat{x}$ carries the specific causal effect the model uses for a task.

Explained variance is dominated by high-variance directions shared by all inputs, while task-relevant differences can be small, low-variance directions. Good reconstruction therefore does not guarantee the other three properties. This paper measures each level separately, at every layer of GPT-2 Small [12].

Contributions.

- A matched-sparsity comparison of ReLU, TopK and JumpReLU SAEs at all 12 layers of GPT-2 Small, with three seeds (§6).
- A four-level information test that separates readout shift from information loss using frozen and re-trained readers, norm-matched noise and PCA references (§7).
- A within-layer analysis showing that reconstruction error predicts behavioural damage but not truth readout, and that earlier pooled correlations were artefacts of mixing layers (§8).
- A shortcut-free persona-instruction task, a pre-registered causal site sweep, and a decomposition of the causal effect into reconstruction, error and interaction parts (§9).
- Pre-registered fine-tuning experiments, including an audit that found our own earlier truth-aware objective to be circular (§10).

Scope of the claims. All in-house SAEs have 3072 latents (4× expansion), target L0 = 128 and are trained on 101,600 OpenWebText tokens. The results describe these SAEs; whether they hold for larger, longer-trained SAEs is tested only partially (§13).

PENDING: Final framing paragraph (Review #8: lead with the audit's negative finding) after IMP-L and M5.

2 Related Work

Sparse dictionary learning for interpretability. Bricken et al. [3] and Cunningham et al. [4] showed that SAEs trained on transformer activations yield many interpretable latents. Gao et al. [5] introduced TopK SAEs and studied scaling; Rajamanoharan et al. [13] introduced JumpReLU SAEs with a learned threshold and a straight-through estimator. Our JumpReLU uses a simpler sigmoid surrogate and is not the published estimator (§5). Public SAE suites exist for GPT-2 Small via SAELens [1] and for Gemma 2 via Gemma Scope [8].

Evaluating SAEs beyond reconstruction. SAEBench [7] evaluates SAEs on a broad set of downstream metrics and finds that proxy metrics such as reconstruction and sparsity do not reliably predict practical usefulness. AxBench [14] finds that simple baselines outperform SAE-based methods for concept detection and steering. Paulo et al. [11] automate the interpretation of SAE features at scale. Our audit is narrower but more controlled: one model, one task family per question, and pre-registered comparisons with explicit noise, PCA and synonym controls.

Training for downstream faithfulness. Braun et al. [2] train SAEs end-to-end with a KL objective on the model's output. Our output-KL fine-tuning (§10) is inspired by this but is not a reproduction: it fine-tunes existing local SAEs for 500 steps and keeps the local reconstruction term.

Truth and honesty representations. Marks and Tegmark [9] find linear truth structure in LLM activations (the Geometry of Truth datasets we use); Zou et al. [15] introduce representation engineering with persona-contrast prompts, from which our original RepE task was derived.

PENDING: Expand to the 15-reference review from PR #16 (Review #8 H3b); add a differentiation statement per work. Every citation must be checked against the original paper before submission.

3 Background: Transformer Activations and SAEs

GPT-2 Small has 12 transformer blocks with a residual stream of width $d = 768$. We study the activation entering block L, `blocks.`L`.hook_resid_pre`, for $L = 0, \dots, 11$; at $L = 0$ this is the token plus position embedding.

An SAE with m latents maps $x \in \mathbb{R}^d$ to

$$z = \sigma\big(W_e(x - b_d) + b_e\big) \in \mathbb{R}^m, \qquad \hat{x} = W_d z + b_d, \qquad e = x - \hat{x}, \tag{1}$$

with decoder columns renormalised to unit norm after every optimisation step. The three activation functions are

$$\text{ReLU:} \quad z = \mathrm{ReLU}(p), \qquad \text{penalty } \textstyle\sum_i z_i, \tag{2}$$

$$\text{TopK:} \quad z = \mathrm{TopK}_k\big(\mathrm{ReLU}(p)\big),\ k = 128, \qquad \text{no penalty}, \tag{3}$$

$$\text{JumpReLU (surrogate):} \quad z = \mathrm{ReLU}(p) \odot H(p - \theta), \qquad \text{penalty } \textstyle\sum_i \sigma\big((p_i - \theta_i)/T\big), \tag{4}$$

where p is the pre-activation, H the step function and θ a learned per-latent threshold whose gradient uses a sigmoid straight-through estimator with $T = 0.1$. The training loss is $\mathrm{MSE}(\hat{x}, x) + c \cdot \text{penalty}$, and a multiplicative controller adjusts c every 25 steps towards a target L0 of 128.

Metrics. Explained variance is $\mathrm{EV} = 1 - \sum \|x - \hat{x}\|^2 / \sum \|x - \bar{x}\|^2$ and $\mathrm{NMSE} = 1 - \mathrm{EV}$; L0 is the mean number of active latents. An AUROC gap is $\mathrm{AUROC}(x) - \mathrm{AUROC}(\hat{x})$ for a given reader. Spliced ΔCE is the increase in next-token cross-entropy when $\hat{x}$ replaces x at every non-BOS position; loss recovered is $(\mathrm{CE}_{\text{mean-abl}} - \mathrm{CE}_{\mathrm{SAE}})/(\mathrm{CE}_{\text{mean-abl}} - \mathrm{CE}_{\mathrm{orig}})$. Causal quantities are defined in §9.

4 Research Questions and Hypotheses

Main question. Can SAE reconstruction of frozen LLM activations be improved while preserving truth-related information and causal behaviour across transformer layers?

SQ1 (reconstruction). How do ReLU, TopK and JumpReLU SAEs compare across the 12 layers at matched sparsity?

SQ2 (information). Which task information survives reconstruction, and is the loss specific to SAEs compared with generic perturbations of the same size and with PCA at the same reconstruction error?

SQ3 (behaviour). How much does splicing the SAE in damage GPT-2's predictions, and does reconstruction error predict that damage between SAEs at the same layer?

SQ4 (causal). Where does GPT-2 carry a persona-instruction effect, and how much of it survives in the SAE reconstruction?

SQ5 (improvement). Can fine-tuning improve faithfulness at matched sparsity, and at which layers?

RepE questions and hypotheses. For the persona-instruction task (RepE-LM) we ask:

R-Q1 (representation). Is the persona information lost in reconstruction, or only misread by a frozen reader? Prior expectation written in the RLM-RP pre-registration: re-trained gaps are much smaller than frozen gaps, as for truth (M2).

R-Q2 (geometry and wording). Is the honest-minus-dishonest difference preserved, and does any preservation transfer to held-out instruction wording?

R-Q3 (causal site and decomposition). Where does GPT-2 carry the effect (M3), and how much of it does the reconstruction carry (M4)?

R-Q4 (improvement). Does Method A or output-KL fine-tuning raise representational (RLM-RP) or causal (E3) persona preservation, relative to the base SAE and an OWT-only control? Hypothesis stated before the runs: Method A does not, because it never sees persona prompts.

That expectation was *not borne out* for persona (§7.1); R-Q4's prior for Method A held, and output-KL gave no creditable gain (§10.2).

Evidential status. Each result is labelled TEST (held-out split, evaluated once under a written protocol), VALIDATION, or PRELIMINARY (a validation pilot whose pre-registered criterion was not met or which is unreplicated). Every experiment from M0 onwards was pre-registered with file hashes before it ran.

The project's original proposal asked whether SAE features are causally faithful to their auto-generated labels. The scope was narrowed to the questions above after instructor reviews and internal audits (Appendix A).

5 Datasets and Experimental Setup

OpenWebText (OWT). 1000 documents sampled from `Skylion007/openwebtext` (streaming shuffle, seed 42, 100–20,000 characters, exact deduplication), split 800/100/100 by document. Each document is truncated to 128 tokens; excluding BOS this gives 101,600 / 12,700 / 12,700 training, validation and test tokens.

Geometry of Truth (GoT). The `cities` and `neg_cities` sets [9] (1496 statements each) form the "geographic" truth task, with city-grouped splits of 1046/224/226 per set. The numerical sets have a surface shortcut (the second number alone gives AUROC 0.84) and are used only as a diagnostic.

RepE-LM persona task. Our original persona prompts ("Pretend you're an honest/untruthful person") had a length shortcut: *untruthful* is three tokens longer than *honest*, and prompt length alone predicts the label with AUROC 0.80. We therefore built RepE-LM: 506 statements (train 417, validation 89) times 20 prompt variants, in which the two prompts of each pair differ in exactly one token at a fixed position. Templates T1–T4 swap truthful/deceptive, trustworthy/dishonest, sincere/lying and facts/lies, each with an honest-synonym control; T5–T6 are held out; N1–N2 swap neutral words (tall/short, red/blue). Seven pre-registered gates (Table 1) verify the construction.

SAE training. For each layer, architecture and seed (42, 1042, 2042) we train one SAE with width 3072, Adam (learning rate 3×10^{-4}), batch 256 sampled with replacement, gradient clipping at 10, and one uninterrupted run of 15,000 steps (3.84M samples, about 38 passes over the training tokens). Architectures are compared only when every run's L0 is within $\pm 10\,\%$ of the target; 107 of 108 runs qualify (layer-0 JumpReLU seed 42 does not).

Splits. Probes and selections use training data; results are reported on validation unless labelled TEST. The RepE-LM test split has never been built or accessed. The GoT test split was scored in earlier frozen-probe evaluations and is treated as previously exposed (Appendix B).

Evidence status of every result in this paper. Apart from the OWT reconstruction table (TEST), *every* result reported here (M0–M4 and the fine-tuning experiments) is a VALIDATION finding. Hypotheses, criteria and scripts were fixed before each run, but the held-out test evaluation (M5) has not been run, so no result is a confirmed held-out finding. When a site or effect selected on training prompts is said to have *replicated*, it means only that it held again on validation prompts. Fine-tuning classes (PRELIMINARY IMPROVEMENT, NO IMPROVEMENT, NOT CREDITABLE, REGRESSION) are validation classes; none of them is a confirmed result.

6 Reconstruction and Sparsity

Table 2 and Figure 1 (left) give OWT test reconstruction for all 108 SAEs (TEST). Explained variance falls with depth for every architecture, from 0.93–0.95 at layers 0–2 to 0.757 (ReLU) and 0.815 (TopK, JumpReLU) at layer 11. Among matched runs, ReLU leads at layer 1, ReLU and JumpReLU tie at layer 2, JumpReLU leads at layers 3–9, and JumpReLU and TopK tie at layers 10–11. Seed standard deviations are ≤ 0.001 except for layer-0 JumpReLU, whose seed 42 is unmatched (L0 73.6; about 96 % of latents dead) and excluded from rankings.

Table 1: M1 RepE-LM construction and gates (pre-registered pass rules). Source: `results/repe/repe_lm_gates/gates.json`.

Gate	Status	Evidence (train statements)
K1 construction	PASS	6,072 pairs; max residual difference before the swapped token 0.0 at all 12 layers
K2 shortcuts	PASS	max deviation of any shortcut AUROC from 0.5: 5.6e-17
K3 positive control	PASS	unmasked bag-of-words AUROC 1.000
K5 persona meaning	PASS	leave-one-pair-out AUROC 0.701 at layer 8 (CI low 0.689); passing layers [8, 11]
K6 leakage	PASS	0 exact overlaps; 0 same-label near-duplicates
K7 behaviour	PASS	median KL(H‖U) 0.0186 vs KL(H‖H′) 0.0092 nats; ratio 1.99; Wilcoxon $p = 3e-68$

Table 2: Reconstruction on the OWT test split (TEST; 12,700 tokens), width 3072, target L0 = 128. Layer-0 JumpReLU includes the unmatched seed 42 (L0 73.6). Source: `results/openwebtext/reconstruction_evaluation/uninterrupted_protocol/all_layers_summary.csv`.

Layer	OWT test EV (mean±sd, 3 seeds)			OWT test L0		
	ReLU	TopK	JumpReLU	ReLU	TopK	JumpReLU
0	0.950±0.000	0.911±0.000	0.826±0.101	135.1	125.3	110.3
1	0.937±0.000	0.921±0.001	0.933±0.000	132.2	128.0	129.3
2	0.936±0.000	0.926±0.000	0.935±0.002	131.9	128.0	129.0
3	0.905±0.000	0.906±0.000	0.915±0.001	131.7	128.0	129.1
4	0.872±0.000	0.884±0.000	0.893±0.001	131.1	128.0	129.1
5	0.848±0.000	0.868±0.000	0.875±0.000	130.8	128.0	129.0
6	0.832±0.001	0.857±0.000	0.865±0.000	131.0	128.0	129.1
7	0.819±0.001	0.850±0.000	0.855±0.000	131.4	128.0	129.9
8	0.805±0.000	0.840±0.000	0.844±0.000	131.9	128.0	130.3
9	0.789±0.001	0.831±0.000	0.833±0.000	132.4	128.0	131.0
10	0.773±0.000	0.822±0.000	0.823±0.000	132.7	128.0	131.6
11	0.757±0.001	0.815±0.000	0.815±0.000	133.2	128.0	132.1

Sparsity is matched on OWT only. ReLU runs 2–6 % above the target L0. A within-layer adjustment using the L0 elasticity measured in an earlier sweep changes no ranking except two near-ties (M0). The architectures differ strongly in effective width: of 3072 latents, ReLU keeps 2920–3063 alive, TopK 1711–2851 and JumpReLU 469–2947. On task prompts the number of active latents is not matched (on GoT final tokens, TopK uses 128, ReLU 32–110 and JumpReLU 23–92 at layers 1–11), which confounds architecture comparisons on task information (§7).

PENDING: E1 (TopK trained at several k to match task-level L0) and E2 (public `gpt2-small-res-jb` reference; not matched in width, data or L0).

7 Information Preservation

All results in this section are VALIDATION findings (M2; 108 SAEs, 36 PCA references).

Frozen probes overstate the loss. A logistic truth probe fitted on original activations and applied unchanged to SAE reconstructions loses 0.28–0.29 AUROC on GoT (Table 3, Figure 2).

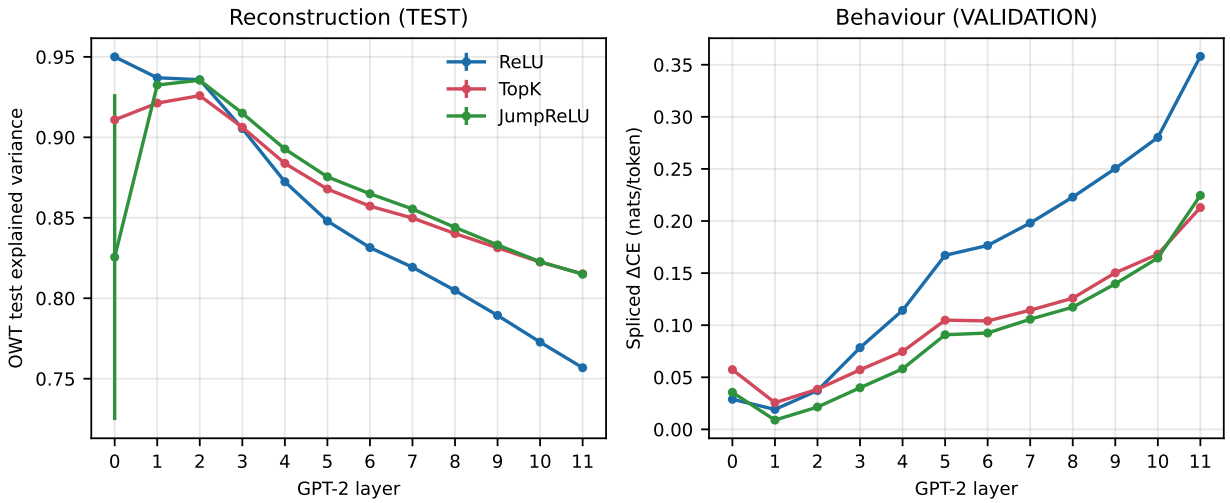


Figure 1: Left: OWT test explained variance by layer (TEST; error bars: seed sd). Right: spliced next-token cross-entropy increase (VALIDATION). Sources: Tables 2 and 6.

Earlier versions of this project read this as truth information being lost. Adding isotropic noise with the same norm as the SAE error causes almost the same drop (0.25–0.26), and a simpler mass-mean reader loses only 0.02–0.03. Most of the frozen-probe collapse is therefore sensitivity of that reader to any perturbation of this size, not SAE-specific loss.

A smaller, real loss remains. Readers re-trained on the reconstructions are 0.07–0.12 AUROC worse than readers trained on the original activations for GoT, and 0.14–0.18 worse for the persona task. Re-training never recovers the original performance, so no cell is classified as "readout shifted"; no cell is "preserved". The task label can be decoded from the reconstruction error alone (AUROC 0.62–0.77), so the lost information is still in the activation, outside the SAE's reconstruction.

SAEs are never better than PCA. At the same OWT reconstruction error, PCA keeps at least as much task information as the SAE in every cell; the SAE is significantly worse in 52 of 81 GoT cells, 59 of 99 persona cells and 81 of 90 neutral cells (one-sided paired bootstrap, BH-corrected).

Persona versus generic change. Part of the persona loss is generic one-word-change information: the re-trained gap is 0.07–0.13 larger for persona than for neutral swaps at layers 3–8, and about equal at layers 10–11.

Architecture differences are confounded. On truth, ReLU loses least and JumpReLU most, the reverse of the reconstruction ranking; on persona, ReLU loses most. Because task-level L0 is not matched, we do not interpret these differences as architecture effects until E1 is run.

7.1 RepE-LM persona preservation in detail (RLM-RP)

All results in this subsection are VALIDATION findings: 216 SAEs, i.e. the 108 base SAEs plus the 81 IMP-L and 27 layer-6 fine-tunes, on the equal-length RepE-LM prompts (final token). Probes and C are fitted on training statements only. The historical length-confounded RepE results are not

Table 3: M2 information preservation (VALIDATION): AUROC gaps (original − reconstructed; ↓), mean over informative layers (GoT 4–11, RepE-LM 1–11) and 3 seeds. "Noise adds isotropic noise with the norm of the SAE error; "PCA is matched to the SAE's OWT validation NMSE. Error AUROC: label decoded from the error term alone. Source: `results/shared/m2_four_level_faithfulness/summary_by_layer_arch.csv`.

Task	SAE	Frozen log.	Noise	Frozen m.-mean	Re-trained log.	PCA re-tr.	MLP	Error AUROC
GoT	ReLU	0.287	0.259	0.033	0.066	0.042	0.049	0.675
GoT	TopK	0.275	0.246	0.024	0.075	0.032	0.037	0.615
GoT	JumpReLU	0.286	0.257	0.026	0.118	0.030	0.066	0.632
Persona	ReLU	0.198	0.076	0.091	0.179	0.131	0.232	0.773
Persona	TopK	0.162	0.072	0.066	0.147	0.116	0.190	0.719
Persona	JumpReLU	0.155	0.064	0.062	0.143	0.114	0.177	0.721
Neutral	ReLU	0.113	0.048	0.049	0.099	0.066	0.033	0.636
Neutral	TopK	0.098	0.046	0.038	0.090	0.064	0.031	0.615
Neutral	JumpReLU	0.094	0.043	0.035	0.083	0.062	0.031	0.619

used. The run reproduces M2's base-SAE values exactly (maximum difference 0.0) and IMP-L's re-trained persona gaps to 1.6×10^{-4}.

Persona information is lost, not only misread. Unlike the GoT case above, the persona frozen-logistic gap (0.10–0.24; Table 4, Figure 3) is 1.9–3.4 times the gap caused by same-norm noise. A logistic reader re-trained on the reconstructions recovers almost none of it: the re-trained gap is 0.08–0.22, and its statement-bootstrap interval excludes zero for all 99 base SAEs at layers 1–11. At layers 2–8 the re-trained persona gap exceeds the neutral-word control by 0.06–0.12, so part of the mid-layer loss is persona-specific. At layers 10–11 it does not exceed the neutral gap, so the late loss is generic.

The honest-minus-dishonest difference is distorted. For each pair, the reconstructed difference $\hat{\Delta} = \hat{x}_H - \hat{x}_U$ has a cosine of only 0.52–0.58 (ReLU), 0.35–0.43 (TopK) and 0.38–0.48 (JumpReLU) with $\Delta = x_H - x_U$. ReLU shrinks the difference (norm ratio 0.65–0.71). TopK and JumpReLU inflate it (1.2–1.5) with an error part larger than Δ itself. Averaged over pairs, the mean difference direction is kept better (cosine 0.70–0.86). This is the representational counterpart of the causal leakage measured in §9.

Wording, shortcuts and robustness.

- **Held-out wording.** The persona probe transfers only weakly to the never-fitted templates T5/T6, even on original activations (AUROC 0.54–0.69). The SAEs lose a further 0.03–0.13.
- **Synonyms.** Honest-synonym prompts behave like the honest prompts: the synonym gap is within 0.024 of the persona gap. The probe barely separates honest from synonym (AUROC 0.52–0.54), so it mostly reads meaning, not the word.
- **Length.** Every pair has equal token length, so the length shortcut of the original prompts cannot contribute.
- **Near duplicates.** Dropping the 14 validation statements that have an opposite-label near duplicate in training changes any gap by at most 0.02.

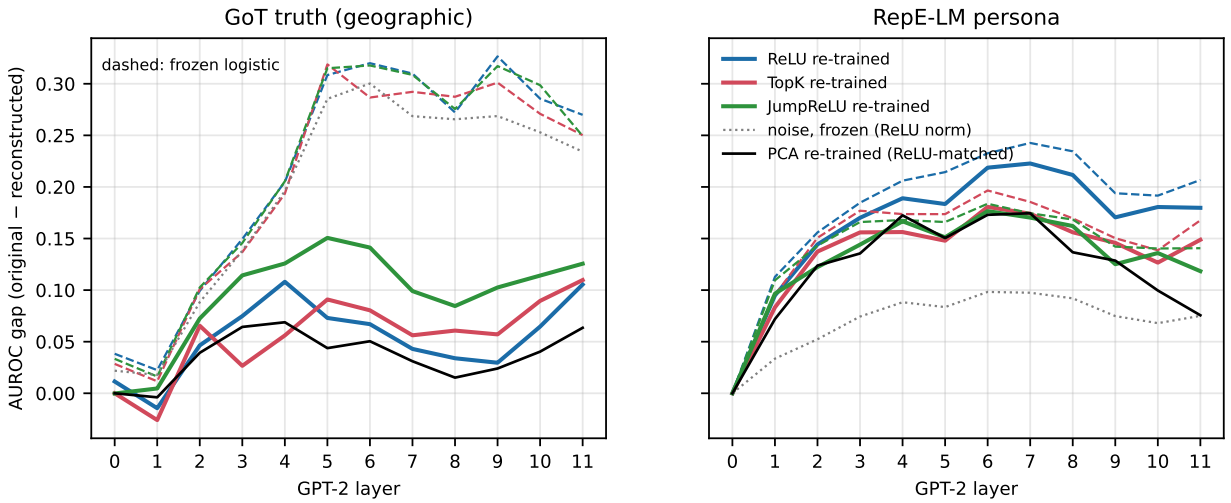


Figure 2: Information preservation by layer (VALIDATION). Dashed: frozen logistic gap; solid: re-trained logistic gap; dotted grey: frozen gap under norm-matched noise; black: PCA at the ReLU SAE's error.

Architecture differences are partly sparsity. TopK and JumpReLU have smaller persona gaps than ReLU at most layers, but on these prompts ReLU and JumpReLU use 142–214 latents per token against TopK's 128. In an exploratory control that keeps each token's 128 largest latents, JumpReLU's re-trained gap rises from 0.144 to 0.210 at layer 3 and from 0.162 to 0.188 at layer 8, while ReLU changes by at most 0.009. We therefore do not rank architectures on persona preservation. A trained task-level L0 match (E1) remains pending.

Fine-tuning barely moves persona preservation. Method A changes neither the frozen nor the re-trained persona gap in any of 12 layer×architecture cells: every difference lies between −0.004 and +0.006 (Table 5). Output-KL meets the pre-registered rule in three cells for each reader. On the frozen reader these are all three architectures at layer 3. On the re-trained reader they are ReLU and TopK at layer 3 and JumpReLU at layer 6. Every output-KL change is at most 0.007 AUROC, against gaps of 0.12–0.24, and for ReLU and JumpReLU it comes with 4–9 % more active latents on these prompts. Four output-KL cells are not creditable (IMP-L L0 guard). No cell regresses.

PENDING: M5 held-out confirmation; E1 task-level sparsity control; E2 external SAE reference.

8 Behavioural Faithfulness

Splicing the reconstruction into GPT-2 at every non-BOS position increases next-token cross-entropy by 0.009–0.026 nats at layer 1 and by 0.213–0.358 nats at layer 11 (Table 6, VALIDATION); loss recovered relative to mean ablation stays at or above 0.93. An identity splice changes nothing (exactly 0). The true-versus-false likelihood preference on GoT `cities` (0.82 originally) is unchanged within ±0.025 through layer 8 and drops by 0.04–0.09 at layers 9–11.

SAE error is more harmful than random error. Replacing the SAE error by isotropic noise of the same norm gives a smaller cross-entropy increase in most runs: the ratio of SAE-induced to

Table 4: RLM-RP: RepE-LM persona preservation by layer for the 99 base SAEs at layers 1–11 (VALIDATION; final token; mean±sd over 3 seeds where shown). "Orig.: frozen logistic AUROC on original activations. Gaps are original − reconstructed AUROC (↓). "Noise: frozen gap under isotropic noise with the SAE error's norm. Held-out: frozen persona probe on never-fitted templates T5/T6. $\Delta = x_H - x_U$ per pair; EV and L0 on all RepE-LM validation prompts (L0 matched on OWT, not here). Layer 0 is excluded: AUROC is 0.5 by construction. Source: `results/repe/repe_lm_representation_preservation/summary_by_layer_arch.csv`; `results/repe/repe_lm_representation_preservation/original_by_layer.csv`.

L	Orig.	SAE	Frozen log. gap	Noise	Frozen m.-m.	Re-trained gap	Held-out T5/T6	Neutral re-tr.	$\cos(\Delta, \hat{\Delta})$	$\|\hat{\Delta}\|/\|\Delta\|$	m.-m. cos	EV	L0
1	0.657	ReLU	0.113±0.002	0.033	0.035	0.095±0.006	0.026	0.085	0.52	0.69	0.74	0.896	214
		TopK	0.095±0.003	0.037	0.029	0.083±0.012	0.026	0.085	0.37	1.46	0.73	0.857	128
		JumpReLU	0.109±0.003	0.035	0.027	0.097±0.004	0.028	0.082	0.38	1.31	0.74	0.909	208
2	0.728	ReLU	0.156±0.004	0.054	0.055	0.144±0.005	0.066	0.072	0.58	0.69	0.77	0.898	201
		TopK	0.151±0.006	0.060	0.055	0.137±0.007	0.060	0.080	0.40	1.33	0.70	0.874	128
		JumpReLU	0.143±0.008	0.049	0.046	0.122±0.009	0.060	0.073	0.43	1.33	0.74	0.915	207
3	0.780	ReLU	0.185±0.006	0.073	0.082	0.170±0.006	0.087	0.064	0.56	0.70	0.77	0.870	189
		TopK	0.177±0.010	0.076	0.073	0.156±0.020	0.077	0.068	0.38	1.32	0.75	0.859	128
		JumpReLU	0.166±0.004	0.063	0.067	0.144±0.011	0.075	0.052	0.43	1.29	0.77	0.901	201
4	0.797	ReLU	0.206±0.007	0.085	0.102	0.189±0.012	0.098	0.059	0.54	0.67	0.74	0.836	176
		TopK	0.174±0.005	0.084	0.072	0.156±0.009	0.064	0.058	0.38	1.36	0.77	0.837	128
		JumpReLU	0.168±0.007	0.071	0.078	0.167±0.008	0.075	0.048	0.42	1.33	0.78	0.880	188
5	0.793	ReLU	0.214±0.010	0.082	0.105	0.183±0.015	0.108	0.076	0.52	0.65	0.72	0.806	171
		TopK	0.174±0.006	0.078	0.076	0.148±0.011	0.085	0.070	0.35	1.41	0.75	0.818	128
		JumpReLU	0.166±0.011	0.068	0.069	0.151±0.010	0.074	0.063	0.40	1.35	0.77	0.858	181
6	0.830	ReLU	0.233±0.020	0.097	0.113	0.219±0.016	0.110	0.086	0.53	0.65	0.74	0.786	164
		TopK	0.197±0.006	0.091	0.080	0.181±0.019	0.092	0.079	0.36	1.43	0.77	0.805	128
		JumpReLU	0.184±0.004	0.079	0.079	0.177±0.007	0.087	0.070	0.41	1.35	0.79	0.846	178
7	0.860	ReLU	0.243±0.012	0.097	0.125	0.223±0.011	0.128	0.092	0.54	0.66	0.75	0.771	163
		TopK	0.185±0.012	0.091	0.083	0.174±0.010	0.100	0.086	0.38	1.40	0.80	0.794	128
		JumpReLU	0.174±0.005	0.080	0.076	0.170±0.021	0.098	0.079	0.42	1.35	0.81	0.830	173
8	0.852	ReLU	0.235±0.004	0.092	0.112	0.212±0.004	0.123	0.089	0.56	0.67	0.76	0.752	161
		TopK	0.170±0.006	0.085	0.073	0.156±0.017	0.096	0.084	0.39	1.38	0.82	0.782	128
		JumpReLU	0.168±0.009	0.077	0.075	0.162±0.004	0.086	0.078	0.44	1.34	0.84	0.811	167
9	0.838	ReLU	0.194±0.010	0.075	0.092	0.171±0.007	0.088	0.113	0.57	0.70	0.77	0.731	163
		TopK	0.150±0.011	0.068	0.067	0.146±0.020	0.065	0.093	0.42	1.38	0.85	0.766	128
		JumpReLU	0.142±0.012	0.063	0.063	0.125±0.003	0.064	0.092	0.46	1.31	0.85	0.789	160
10	0.831	ReLU	0.192±0.005	0.068	0.085	0.181±0.005	0.072	0.159	0.57	0.71	0.79	0.707	163
		TopK	0.139±0.007	0.060	0.059	0.127±0.006	0.055	0.129	0.43	1.35	0.85	0.753	128
		JumpReLU	0.140±0.011	0.058	0.057	0.136±0.007	0.056	0.126	0.46	1.28	0.85	0.763	149
11	0.865	ReLU	0.207±0.006	0.075	0.090	0.180±0.021	0.072	0.198	0.56	0.71	0.80	0.659	160
		TopK	0.168±0.014	0.064	0.063	0.149±0.015	0.057	0.154	0.43	1.35	0.85	0.729	128
		JumpReLU	0.141±0.014	0.064	0.047	0.118±0.018	0.048	0.150	0.48	1.22	0.86	0.732	142

noise-induced damage is 0.76–2.89 per run (median 1.64); 7 of 107 runs are below 1, all at layers 1–3.

Within a layer, reconstruction error predicts behavioural damage but not truth readout (M0). Table 7 reports within-layer regressions over the 107 matched SAEs. Lower OWT NMSE goes with less splice damage ($\beta = 0.165$ per unit log NMSE, Holm $p = 0.0008$; per-layer Spearman ρ 0.80–0.97 from layer 3). For the frozen GoT truth gap, the formal test passes (Holm $p = 0.044$), but architecture explains 99.8 % of the within-layer NMSE variance and the association disappears without architecture fixed effects ($p = 0.89$); GoT reconstruction error has no association. Two earlier claims are corrected: a pooled correlation of 0.89 between reconstruction and persona preservation is 0.05 within layer, and the range of the SAE-versus-noise damage ratio is the one given above.

Table 5: RLM-RP: effect of fine-tuning on RepE-LM persona gaps (VALIDATION; change in AUROC gap, ↓ is better; mean [95% t-interval] over 3 seeds) and pre-registered classes for the frozen and re-trained reader separately. "not cred.": an IMP-L OWT guard failed. *: RepE-LM L0 outside ±5% of base on some seed (reported, not a guard). ‡ layer 6: no IMP-L-format guards exist, so classes there are not guard-checked. Source: `results/repe/repe_lm_representation_preservation/comparisons.csv`; `results/repe/repe_lm_representation_preservation/criteria.json`.

L	SAE	A−base frozen	A−base re-trained	A class (fr./re.)	KL−A frozen	KL−A re-trained	KL class (fr./re.)	KL RepE L0 Δ%
3	ReLU	+0.0001 [-0.0027, +0.0028]	-0.0000 [-0.0070, +0.0070]	no impr. / no impr.	-0.0037 [-0.0054, -0.0021]	-0.0030 [-0.0062, +0.0002]	PRELIM. / PRELIM.	+4.6
	TopK	+0.0035 [-0.0034, +0.0103]	+0.0038 [-0.0031, +0.0106]	no impr. / no impr.	-0.0043 [-0.0091, +0.0006]	-0.0060 [-0.0082, -0.0037]	PRELIM. / PRELIM.	+0.0
	JumpReLU	-0.0020 [-0.0041, +0.0000]	+0.0033 [-0.0013, +0.0079]	no impr. / no impr.	-0.0056 [-0.0123, +0.0010]	-0.0068 [-0.0160, +0.0023]	PRELIM. / no impr.	+4.3
6‡	ReLU	-0.0020 [-0.0053, +0.0013]	-0.0017 [-0.0046, +0.0012]	no impr. / no impr.	+0.0005 [-0.0005, +0.0015]	-0.0007 [-0.0028, +0.0014]	no impr. / no impr.	+5.7*
	TopK	-0.0006 [-0.0173, +0.0160]	+0.0058 [-0.0169, +0.0286]	no impr. / no impr.	+0.0001 [-0.0080, +0.0081]	-0.0040 [-0.0187, +0.0106]	no impr. / no impr.	+0.0
	JumpReLU	-0.0020 [-0.0048, +0.0008]	-0.0035 [-0.0099, +0.0029]	no impr. / no impr.	-0.0052 [-0.0135, +0.0032]	-0.0055 [-0.0097, -0.0012]	no impr. / PRELIM.	+6.4*
8	ReLU	+0.0005 [-0.0045, +0.0054]	+0.0003 [-0.0014, +0.0020]	no impr. / no impr.	-0.0038 [-0.0065, -0.0012]	-0.0043 [-0.0080, -0.0006]	not cred. / not cred.	+6.2*
	TopK	-0.0031 [-0.0093, +0.0031]	-0.0036 [-0.0115, +0.0044]	no impr. / no impr.	+0.0011 [-0.0123, +0.0145]	+0.0020 [-0.0071, +0.0111]	no impr. / no impr.	+0.0
	JumpReLU	-0.0018 [-0.0071, +0.0034]	-0.0004 [-0.0128, +0.0120]	no impr. / no impr.	-0.0045 [-0.0107, +0.0018]	-0.0036 [-0.0142, +0.0070]	not cred. / not cred.	+8.1*
9	ReLU	-0.0022 [-0.0048, +0.0004]	-0.0011 [-0.0049, +0.0027]	no impr. / no impr.	+0.0001 [-0.0025, +0.0027]	-0.0021 [-0.0058, +0.0015]	not cred. / not cred.	+5.8*
	TopK	-0.0014 [-0.0059, +0.0030]	-0.0003 [-0.0031, +0.0026]	no impr. / no impr.	-0.0003 [-0.0092, +0.0086]	-0.0021 [-0.0070, +0.0028]	no impr. / no impr.	+0.0
	JumpReLU	-0.0025 [-0.0080, +0.0031]	-0.0029 [-0.0142, +0.0083]	no impr. / no impr.	-0.0067 [-0.0300, +0.0166]	-0.0029 [-0.0220, +0.0161]	not cred. / not cred.	+8.6*

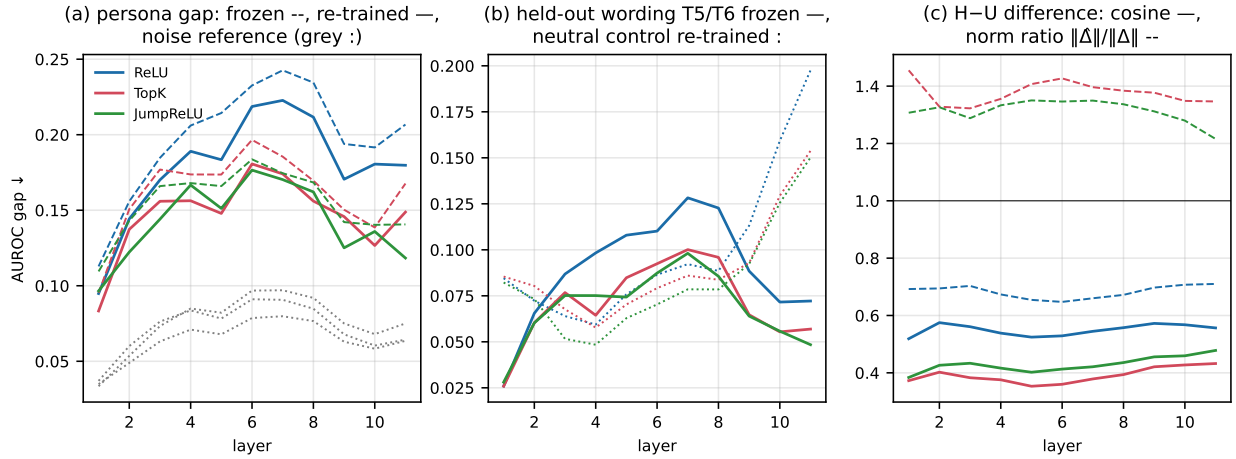


Figure 3: RLM-RP (VALIDATION; 99 base SAEs, mean over 3 seeds). (a) Persona gaps: frozen logistic, re-trained logistic, and the same-norm noise reference. (b) Frozen gap on held-out wording T5/T6, and the re-trained gap on the neutral control. (c) Per-pair cosine and norm ratio between the reconstructed and original H−U difference.

9 Causal Faithfulness

All results in this section are VALIDATION findings (M3 sites selected on training prompts and replicated on validation prompts; M4 on validation prompts).

Where GPT-2 carries the persona effect (M3). For each RepE-LM pair we run GPT-2 on the honest (H) and the dishonest (U) prompt and measure $\mathrm{KL}(p_H \| p_U)$ on the next token (mean 0.033 nats). Copying the honest activation into the dishonest run at one layer and position group restores a share E of that divergence. Figure 4 shows that the effect is carried by the persona word early (layer 1: $E = 0.97$) and by the final token late (layer 11: $E = 0.95$), with the handover at layers 7–8; the instruction words after the persona slot carry up to 0.28. Of 65 cells, 50 pass the pre-registered test (Holm-corrected, above the 95th percentile of 200 matched-norm random vectors). The pre-registered rule selected the layer-1 persona slot and the layer-11 final token, both replicated

Table 6: Behavioural faithfulness: GPT-2 with the SAE reconstruction spliced in at every non-BOS position (OWT validation; original CE 3.396 nats/token; matched seeds only). Source: `results/openwebtext/spliced_evaluation/summary.csv`.

Layer	Spliced ΔCE (nats/token, ↓)			Loss recovered (↑)		
	ReLU	TopK	JumpReLU	ReLU	TopK	JumpReLU
0	0.029	0.057	0.036	0.996	0.991	0.995
1	0.019	0.026	0.009	0.998	0.997	0.999
2	0.037	0.038	0.021	0.995	0.994	0.997
3	0.078	0.057	0.040	0.987	0.991	0.994
4	0.114	0.075	0.058	0.984	0.990	0.992
5	0.167	0.105	0.091	0.967	0.980	0.982
6	0.177	0.104	0.092	0.966	0.980	0.982
7	0.198	0.114	0.106	0.963	0.979	0.980
8	0.223	0.126	0.117	0.958	0.976	0.978
9	0.250	0.150	0.140	0.951	0.971	0.973
10	0.280	0.168	0.164	0.947	0.968	0.969
11	0.358	0.213	0.225	0.930	0.958	0.956

Table 7: M0 within-layer regressions (VALIDATION): outcome $\sim \beta\cdot$predictor + layer FE + architecture FE + log L0, Freedman–Lane permutation within layer. The GoT row passes formally but vanishes without architecture fixed effects ($p = 0.89$; see text). Source: `results/shared/m0_within_layer_reanalysis/within_layer_regressions.csv`.

Predictor → outcome	β	95% CI	p_{perm}	p_{Holm}	n
log OWT NMSE → GoT frozen-logistic gap	0.268	[0.056, 0.477]	0.0146	0.0438	72
log GoT relative error → GoT frozen-logistic gap	0.005	[-0.059, 0.080]	0.8680	1.0000	72
log OWT NMSE → OWT spliced ΔCE	0.165	[0.134, 0.265]	0.0002	0.0008	107
log GoT relative error → OWT spliced ΔCE	-0.012	[-0.060, 0.028]	0.6267	1.0000	107

on validation prompts; a pre-registered amendment added the layer-8 final token (train $E = 0.505$), replicated on validation without SAEs. A same-meaning synonym swap travels the same route at about half the size. Template T1 (truthful/deceptive) has no effect beyond its synonym control and serves as a control case; T2–T4 are the primary set.

How much of the effect the SAE carries (M4). At a site, let $\Delta = x_H - x_U$, $\Delta_{\text{rec}} = \hat{x}_H - \hat{x}_U$ and $\Delta_{\text{err}} = e_H - e_U$, so $\Delta = \Delta_{\text{rec}} + \Delta_{\text{err}}$ exactly. With $T(v) = \text{KL}(p_H \| p_U) - \text{KL}(p_H \| p_{U+v})$, we define $T_{\text{full}} = T(\Delta)$, $T_{\text{rec}} = T(\Delta_{\text{rec}})$, $T_{\text{err}} = T(\Delta_{\text{err}})$, the interaction $I = T_{\text{full}} - T_{\text{rec}} - T_{\text{err}}$, and Shapley shares $\phi_{\text{rec}} = (T_{\text{rec}} + T_{\text{full}} - T_{\text{err}})/2$. The patch adds v to the dishonest run's original activation; it is not a full splice. All exact checks pass (Δ identity error 9.5×10^{-7}; T_{full} reproduces M3 to 7×10^{-8}).

Table 8 and Figure 5 show that the reconstructed difference restores 93–97 % of the effect at the layer-1 persona token, but only 62–68 % at the layer-8 and layer-11 final token. There, ReLU's error part alone carries 40–45 %; for TopK and JumpReLU, the interaction is large ($|I|/T_{\text{full}}$ 0.26–0.38), so their shares cannot be read additively. The persona-minus-synonym difference is preserved in the reconstruction at every site. Higher explained variance does not imply higher causal preservation: at layer 11, TopK has higher EV than ReLU (0.815 vs 0.757) but keeps less of the effect (0.62 vs 0.68).

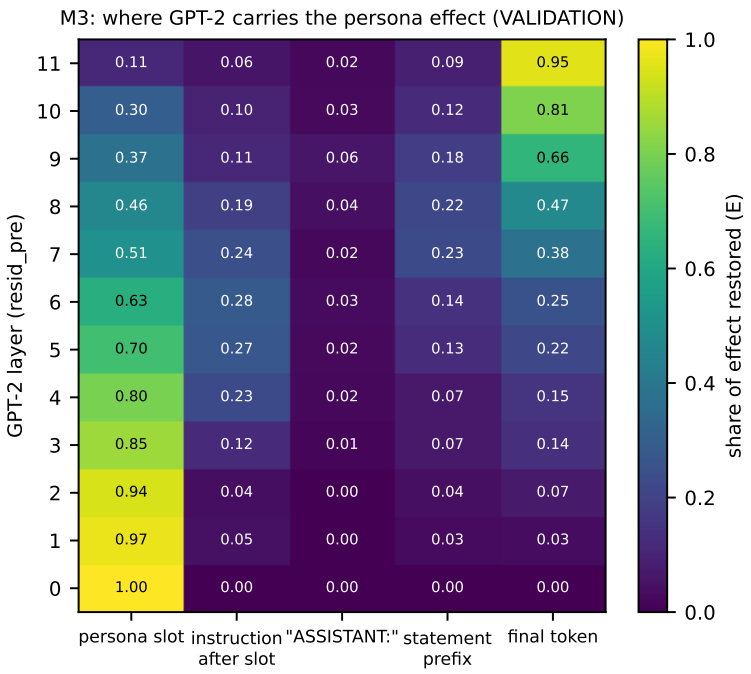


Figure 4: M3 (VALIDATION): share of the persona-instruction effect restored by patching one position group at one layer of GPT-2 Small.

SAE-in-the-loop feature patches. Encoding both prompts, setting the dishonest prompt's top-k features to their honest values, decoding and keeping the error term restores more than 200 frequency- and magnitude-matched random feature sets in all 27 SAE $\times$ site combinations ($p = 0.005$, the minimum attainable). The effect is distributed: one feature carries 2–10 % of T_{rec} and 64 features 47–74 %. This shows that SAE-defined directions are causally sufficient when inserted; it does not show that GPT-2 uses SAE features.

Whether fine-tuning changes this decomposition is tested in E3 (§10.2).

PENDING: Minimum detectable effect statement (Review #8 critique 3); M5 held-out confirmation.

10 Fine-tuning Experiments

All results in this section are VALIDATION findings; no fine-tuning result is confirmed.

All fine-tuning starts from a trained SAE and runs 500 Adam steps (learning rate 10^{-5}) with the sparsity controller continued from the base coefficient; ReLU and JumpReLU must stay within ± 5 % of their base L0. Truth is measured with probes fitted on original activations of GoT training groups B, which no training objective sees.

A circular objective, found and corrected. A truth-aware objective that distilled a linear probe's logits appeared to remove most of the truth gap. An audit found that the evaluation probe was the training teacher (same rows, labels and weights), so the result was circular. In a corrected design (teacher fitted on groups A, probes on B, random-teacher control) the truth-aware objective is worse than plain reconstruction fine-tuning on every seed (TopK B-MLP gap 0.189 vs 0.107; random teacher 0.165).

Table 8: M4 causal decomposition (VALIDATION; templates T2–T4; mean±sd over 3 seeds). Shapley shares in parentheses are not interpretable as additive (pre-registered rule: interaction too large). Source: `results/repe/sae_patch_decomposition/architecture_summary.csv`.

Site	SAE	T_{full} (nats)	$T_{\text{rec}}/T_{\text{full}}$	$T_{\text{err}}/T_{\text{full}}$	$\lvert I\rvert/T_{\text{full}}$	ϕ_{rec}
L1 persona slot	ReLU	0.0435	0.938±0.019	0.14	0.12	0.90
	TopK	0.0435	0.931±0.014	0.16	0.14	0.89
	JumpReLU	0.0435	0.968±0.011	0.03	0.10	0.97
L8 final token	ReLU	0.0220	0.621±0.015	0.45	0.12	0.59
	TopK	0.0220	0.625±0.039	0.09	0.32	(0.77)
	JumpReLU	0.0220	0.665±0.011	0.11	0.26	(0.78)
L11 final token	ReLU	0.0430	0.677±0.014	0.40	0.16	0.64
	TopK	0.0430	0.620±0.004	0.06	0.38	(0.78)
	JumpReLU	0.0430	0.678±0.007	0.11	0.28	(0.78)

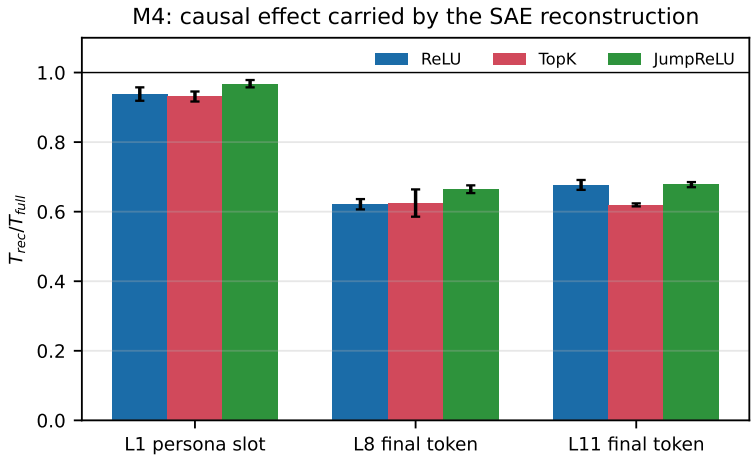


Figure 5: M4 (VALIDATION): share of the causal effect carried by the SAE-reconstructed activation difference ($T_{\text{rec}}/T_{\text{full}}$; mean±sd over 3 seeds; T2–T4).

Method A and output-KL at layer 6. Table 9 reports Method A (GoT reconstruction with OWT replay) and the output-KL variant. Method A cuts GoT reconstruction error by 72–82 % and lowers the independent MLP truth gap on 18 of 18 paired comparisons against both the base and an OWT-only control, but never by more than 2 sd, the pre-registered criterion (PRELIMINARY). Adding output KL lowers OWT spliced KL (0.115→0.102) and GoT statement KL (0.110→0.038) for TopK, meeting its behavioural criterion without changing the truth gap; for ReLU and JumpReLU the KL term pushes L0 5–11 % above the base, so their gains are not creditable. A KL-weight ablation found no truth trade-off to optimise (INCONCLUSIVE).

10.1 Multi-layer improvement experiment (IMP-L)

To test whether the layer-6 findings generalise, we repeated both methods at layers 3, 8 and 9 with an OWT-only control (500 steps of continued training on OWT alone), for all three architectures and seeds: 81 new fine-tunes, evaluated on validation data only. Configuration, criteria and guards were fixed and hashed before any run (`results/geometry_of_truth/multilayer_finetune/preregistration.json`); they are those of layer 6. A criterion counts as met ("consistent") when

Table 9: Layer-6 fine-tuning (VALIDATION; mean over 3 seeds). Method A lowered the B-MLP gap on 18/18 paired comparisons but never by more than 2 sd (pre-registered criterion not met). Output-KL rows for ReLU and JumpReLU fail the ±5% L0 guard and are not creditable. Source: `results/geometry_of_truth/method_a_layer6/summary.csv`; `results/geometry_of_truth/e2e_inspired_layer6/summary.csv`.

SAE	Variant	GoT B-MLP gap ↓	GoT rel. error ↓	OWT EV ↑	OWT ΔCE ↓	OWT L0
ReLU	Base	0.321±0.022	0.059	0.828	0.177	134
	OWT-only control	0.319±0.022	0.058	0.829	0.175	134
	Method A	0.245±0.053	0.017	0.828	0.175	134
	Output-KL (B)	0.280±0.020	0.024	0.830	0.145	141
TopK	Base	0.203±0.030	0.034	0.853	0.104	128
	OWT-only control	0.206±0.035	0.033	0.855	0.102	128
	Method A	0.107±0.020	0.006	0.854	0.102	128
	Output-KL (B)	0.114±0.022	0.009	0.852	0.091	128
JumpReLU	Base	0.292±0.026	0.052	0.862	0.092	131
	OWT-only control	0.287±0.027	0.052	0.864	0.091	131
	Method A	0.238±0.056	0.015	0.863	0.091	131
	Output-KL (B)	0.281±0.046	0.022	0.865	0.076	140

the arm is better on all three seeds and the mean paired difference exceeds twice its standard deviation. Guards require OWT L0 within ±5 % of the base SAE, OWT explained variance no more than 0.005 below the base (output-KL: 0.02 below Method A), spliced ΔCE at most 1.10× the base, and no loss of cities truth preference beyond 0.018. All arms are VALIDATION findings; the 95 % intervals over three seeds are wide by construction ($t_{0.975,2} = 4.30$). Table 12 gives the pre-registered classifications and Appendix Tables 15–17 all paired differences with 95 % intervals; Figure 6 shows the per-arm means.

Method A. Method A cut GoT geographic reconstruction error by 61–85 % in every cell, passed every guard, and lowered the independent truth gap (B-group MLP probe fitted on original activations) relative to the base SAE on all three seeds in all nine layer–architecture cells (mean reduction 4–47 %). The OWT-only control changed the gap by at most 0.021 in either direction, so the reduction is attributable to the GoT data. The pre-registered criterion (consistent against both base and control) was met in four of nine cells: TopK at layer 3, ReLU at layer 8, and ReLU and JumpReLU at layer 9 (PRELIMINARY IMPROVEMENT). In the other five the direction was the same but seed variation exceeded the 2 sd rule (NO IMPROVEMENT under the criterion). Layer 3 counts as truth-informative under the pre-registered rule (original B-MLP AUROC 0.815), although the original logistic probe there reaches only 0.60.

A frozen-probe gain that re-trained readers do not show. A metric reported but not pre-registered as a criterion qualifies this result: the M2-style re-trained logistic gap on GoT geographic statements, and on RepE-LM persona prompts, did not change consistently under Method A relative to the base in any cell (no cell meets the 2 sd rule; all 95 % intervals include zero). Against the OWT-only control, one cell (TopK, layer 8, geographic) is consistently lower. In the four cells where the frozen-probe criterion was met, the re-trained geographic gap changed by −0.005 to +0.010 (three of four upward; ReLU at layer 9 upward on all three seeds). Method A therefore mostly restores what the frozen original-activation probe reads, without measurably restoring the information a re-fitted reader can recover. This mirrors the M2 finding that frozen-probe gaps are dominated by probe

Table 10: IMP-L, Method A − base for four readers (VALIDATION; gap change, ↓ is better; mean [95% t-interval] over 3 seeds). Frozen readers are fitted once on original activations of GoT group B; re-trained readers are re-fitted on the reconstructions. ✓: consistent (lower on all 3 seeds and |mean| > 2 sd) against the base. Only the frozen MLP column is the pre-registered criterion. Source: `results/geometry_of_truth/multilayer_finetune/comparisons.csv`.

Layer	SAE	GoT frozen MLP	GoT frozen logistic	GoT re-trained logistic	Persona re-trained logistic
3	ReLU	-0.018 [-0.041, +0.005]	-0.000 [-0.011, +0.010]	+0.010 [-0.028, +0.048]	-0.000 [-0.007, +0.007]
	TopK	-0.098 [-0.107, -0.089]✓	+0.007 [-0.047, +0.062]	-0.005 [-0.104, +0.094]	+0.004 [-0.003, +0.011]
	JumpReLU	-0.012 [-0.043, +0.019]	+0.005 [-0.013, +0.023]	+0.002 [-0.026, +0.031]	+0.003 [-0.001, +0.008]
8	ReLU	-0.054 [-0.095, -0.013]✓	-0.003 [-0.061, +0.055]	+0.006 [-0.024, +0.035]	+0.000 [-0.002, +0.002]
	TopK	-0.074 [-0.202, +0.054]	-0.045 [-0.130, +0.041]	-0.032 [-0.082, +0.019]	-0.004 [-0.012, +0.004]
	JumpReLU	-0.060 [-0.247, +0.127]	-0.011 [-0.034, +0.013]	-0.005 [-0.023, +0.014]	-0.000 [-0.013, +0.012]
9	ReLU	-0.047 [-0.065, -0.030]✓	+0.007 [-0.067, +0.080]	+0.010 [-0.011, +0.031]	-0.001 [-0.005, +0.003]
	TopK	-0.033 [-0.061, -0.005]✓	-0.007 [-0.094, +0.080]	+0.005 [-0.034, +0.043]	-0.000 [-0.003, +0.003]
	JumpReLU	-0.106 [-0.162, -0.050]✓	-0.002 [-0.048, +0.044]	+0.002 [-0.039, +0.043]	-0.003 [-0.014, +0.008]
consistent cells		5/9	0/9	0/9	0/9

fragility (§7), and it means the Method A truth improvements should not be read as information recovery (DESCRIPTIVE).

Four readers, kept separate. Table 10 reports Method A − base for four different readers, which measure different things:

- the *frozen nonlinear MLP probe*, the pre-registered criterion;
- the *frozen logistic probe*, fitted on the same original group-B activations;
- a *re-trained logistic probe* on GoT;
- a *re-trained logistic probe* on RepE-LM persona prompts.

Against the base alone, the frozen MLP gap falls consistently in 5 of 9 cells; the criterion also requires consistency against OWT-only, which leaves 4 of 9. The frozen logistic gap does not change consistently in any cell, and neither re-trained reader does. The frozen-probe gain is therefore specific to one nonlinear reader fitted on original activations. It is evidence neither of information preservation, which re-trained readers measure, nor of behavioural faithfulness (splicing) or causal faithfulness (patching, §10.2).

Sparsity on every dataset. The pre-registered L0 guard is defined on OWT. Table 11 reports L0 separately on OWT, GoT and RepE-LM prompts.

- **Method A** keeps OWT and RepE-LM L0 within 0.6 %, but makes ReLU and JumpReLU *sparser on GoT statements*: −9.0 to −20.5 % on geographic and −7.5 to −27.5 % on numerical statements. Its truth gains are therefore obtained at lower, not higher, task L0.
- **Output-KL** raises ReLU and JumpReLU L0 on OWT (+3.5 to +8.1 %) and on RepE-LM (+4.3 to +8.6 %) while lowering it on GoT (−5.5 to −29.5 %).
- **TopK** is fixed at 128 on every dataset.

Failed guards are marked in the table. Their cells are excluded from every ranking but are not deleted.

Table 11: IMP-L sparsity by dataset (VALIDATION; mean over 3 seeds): base L0 and the percentage change of each fine-tune, in the order OWT / GoT geographic / GoT numerical / RepE-LM persona prompts. Bold: $|\Delta| > 5\%$. The pre-registered guard is on OWT only; *: output-KL failed it (excluded from rankings). TopK is fixed at 128 on every dataset. Source: `results/geometry_of_truth/multilayer_finetune/comparisons.csv`; `results/geometry_of_truth/multilayer_finetune/summary.csv`; `results/geometry_of_truth/multilayer_finetune/criteria.json`.

Layer	SAE	Base L0 (OWT / GoT geo. / GoT num. / RepE-LM)	OWT-only Δ%	Method A Δ%	Output-KL Δ%
3	ReLU	134 / 45 / 56 / 189	-0.2 / -2.2 / +0.0 / -0.3	-0.2 / **-15.3** / **-27.5** / -0.2	+3.5 / **-16.8** / **-25.3** / +4.6
	TopK	128 / 128 / 128 / 128	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0
	JumpReLU	131 / 29 / 32 / 200	+0.0 / -0.1 / -0.6 / +0.1	+0.1 / **-9.0** / **-13.7** / +0.1	+4.6 / **-5.5** / **-9.3** / +4.3
8	ReLU	134 / 77 / 97 / 161	+0.1 / +1.7 / +0.9 / +0.0	+0.1 / **-13.0** / **-17.1** / -0.1	**+5.8** / **-22.6** / **-27.8** / **+6.2***
	TopK	128 / 128 / 128 / 128	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0
	JumpReLU	132 / 63 / 63 / 167	+0.4 / +0.5 / +0.4 / +0.4	+0.3 / **-9.1** / **-7.5** / +0.5	**+8.1** / **-14.3** / **-16.1** / **+8.1***
9	ReLU	135 / 110 / 90 / 162	+0.2 / +0.6 / +1.4 / +0.1	+0.1 / **-20.5** / **-11.7** / +0.1	**+5.3** / **-29.5** / **-25.4** / **+5.8***
	TopK	128 / 128 / 128 / 128	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0	+0.0 / +0.0 / +0.0 / +0.0
	JumpReLU	132 / 71 / 66 / 160	+0.4 / +0.7 / +1.0 / +0.5	+0.3 / **-13.2** / **-8.8** / +0.6	**+8.1** / **-17.9** / **-12.6** / **+8.6***

Table 12: IMP-L pre-registered classifications (VALIDATION; 3 seeds). "Consistent" requires improvement on all 3 seeds and a mean paired difference larger than 2 sd; NOT CREDITABLE means a guard failed. † original truth probe below the 0.70 informative threshold, so truth results are descriptive. Source: `results/geometry_of_truth/multilayer_finetune/criteria.json`.

Layer	SAE	Method A: truth	failed guards	Output-KL: behaviour	Output-KL: truth	failed guards
3	ReLU	NO IMPROVEMENT	–	PRELIMINARY IMPROVEMENT	REGRESSION	–
	TopK	PRELIMINARY IMPROVEMENT	–	PRELIMINARY IMPROVEMENT	REGRESSION	–
	JumpReLU	NO IMPROVEMENT	–	PRELIMINARY IMPROVEMENT	NO IMPROVEMENT	–
8	ReLU	PRELIMINARY IMPROVEMENT	–	NOT CREDITABLE (guard failed)	NOT CREDITABLE (guard failed)	L0 ±5%
	TopK	NO IMPROVEMENT	–	PRELIMINARY IMPROVEMENT	NO IMPROVEMENT	–
	JumpReLU	NO IMPROVEMENT	–	NOT CREDITABLE (guard failed)	NOT CREDITABLE (guard failed)	L0 ±5%
9	ReLU	PRELIMINARY IMPROVEMENT	–	NOT CREDITABLE (guard failed)	NOT CREDITABLE (guard failed)	L0 ±5%
	TopK	NO IMPROVEMENT	–	PRELIMINARY IMPROVEMENT	NO IMPROVEMENT	–
	JumpReLU	PRELIMINARY IMPROVEMENT	–	NOT CREDITABLE (guard failed)	NOT CREDITABLE (guard failed)	L0 ±5%

Output-KL. Adding the output-KL term lowered OWT spliced KL by 8–18 % and GoT statement KL substantially relative to Method A, consistently on every seed in every cell, and lowered spliced ΔCE by 4–19 %. But for ReLU and JumpReLU it also raised L0: by 3.5 % and 4.6 % at layer 3 (within the guard) and by 5.3–8.1 % at layers 8 and 9 (outside it). The behavioural criterion is therefore met (PRELIMINARY IMPROVEMENT) for all three architectures at layer 3 and for TopK, whose L0 is fixed by construction, at layers 8 and 9; the four ReLU and JumpReLU cells at layers 8 and 9 are NOT CREDITABLE and excluded from rankings. Output-KL never improved the truth gap over Method A. It was consistently worse at layer 3 for ReLU and TopK (REGRESSION under the 2 sd rule, by +0.012 and +0.027; the 95 % intervals just reach zero) and also worse on all seeds at layer 9 for ReLU, a cell classified not creditable because of its L0 guard.

Summary. Across three further layers the layer-6 pattern recurs: task-data fine-tuning reliably lowers task reconstruction error and nudges the frozen truth gap down, and an output-KL term reliably improves behavioural faithfulness, but neither improves truth information as measured by a re-trained reader. For ReLU and JumpReLU at deeper layers the behavioural gain comes with a sparsity cost that the pre-registered ±5 % guard does not allow.

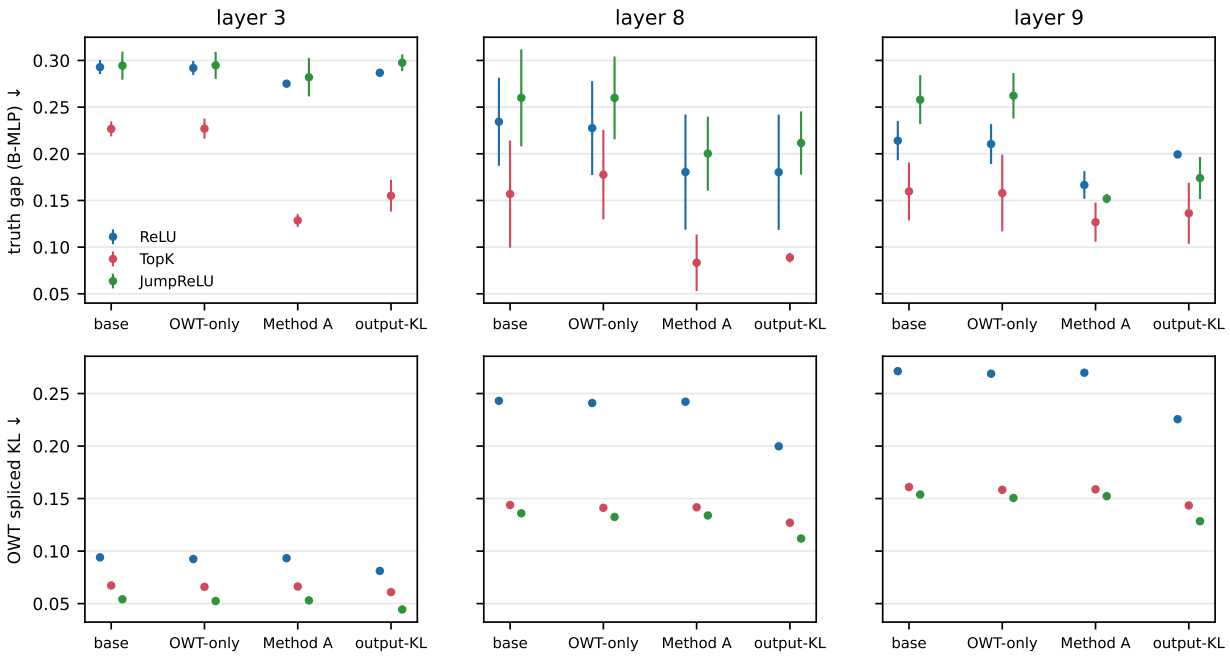


Figure 6: IMP-L per-arm means (±1 sd over 3 seeds) of the independent truth gap (top) and OWT spliced KL (bottom) at layers 3, 8 and 9 (VALIDATION). Output-KL points for ReLU and JumpReLU at layers 8 and 9 violate the pre-registered L0 guard. Source: `results/geometry_of_truth/multilayer_finetune/summary.csv`.

10.2 Causal preservation after fine-tuning (E3)

All results in this subsection are VALIDATION findings. E3 asks whether either fine-tuning method raises the share of GPT-2's persona-instruction effect carried by the SAE reconstruction at the layer-8 final token, the mid-depth site of §9. It applies the M4 decomposition, with M4's code imported unchanged, to 36 SAEs: the 9 base SAEs and the 27 IMP-L layer-8 fine-tunes (OWT-only, Method A, output-KL). No training was involved.

Design. The reconstructed activation difference is added inside GPT-2's forward pass at the site; this is not a steering direction. Hypotheses, comparisons and classes were pre-registered on 2026-11-09 and hashed with all 36 checkpoints before the run:

- **E3-H1:** Method A raises $T_{\text{rec}}/T_{\text{full}}$ consistently against both base and OWT-only.
- **E3-H2:** output-KL raises it over Method A.
- **E3-H3:** any gain must also appear in the persona-minus-synonym restoration.

The output-KL ReLU and JumpReLU SAEs had already failed the IMP-L L0 guard, so their E3-H2 class was NOT CREDITABLE before the run.

Checks. The 9 base SAEs reproduce M4 exactly (maximum difference 0.0, checked before any fine-tuned SAE was evaluated). $T_{\text{full}} = 0.0220$ nats is identical for all 36 SAEs. GPT-2's parameters are unchanged, every patch is at the pair's final token, and all 36 SAEs pass the M4 control tests:

Table 13: E3: causal preservation at the layer-8 final token for the base and IMP-L fine-tuned SAEs (VALIDATION; RepE-LM T2–T4; $T_{\text{full}} = 0.0220$ nats, identical for all 36 SAEs; mean±sd over 3 seeds). $T_{\text{rec}} - S_{\text{rec}}$: persona minus synonym restoration (specificity). H4a/H4c: seeds passing the M4 control tests (reconstruction above matched-norm and decoder-shuffle vectors; top-k^* features above 200 matched random sets). Site EV and L0 on RepE-LM validation prompts. Source: `results/repe/e3_finetuned_causal_preservation/hypotheses.csv`; `results/repe/e3_finetuned_causal_preservation/site_reconstruction.csv`.

SAE	Arm	$T_{\text{rec}}/T_{\text{full}}$	per seed (42, 1042, 2042)	T_{rec} (nats)	$T_{\text{err}}/T_{\text{full}}$	$\lvert I\rvert/T_{\text{full}}$	$T_{\text{rec}} - S_{\text{rec}}$	H4a/H4c seeds	site EV	site L0
ReLU	base	0.621±0.015	(0.630, 0.604, 0.630)	0.0137±0.0003	0.450	0.117	+0.0062	3/3	0.752	161
	OWT-only	0.621±0.016	(0.627, 0.603, 0.633)	0.0137±0.0004	0.450	0.117	+0.0062	3/3	0.753	161
	Method A	0.622±0.016	(0.623, 0.605, 0.636)	0.0137±0.0003	0.449	0.117	+0.0063	3/3	0.752	161
	output-KL	0.640±0.016	(0.645, 0.621, 0.653)	0.0141±0.0004	0.426	0.114	+0.0065	3/3	0.757	171
TopK	base	0.625±0.039	(0.580, 0.650, 0.645)	0.0138±0.0009	0.094	0.323	+0.0070	3/3	0.781	128
	OWT-only	0.632±0.024	(0.604, 0.650, 0.640)	0.0139±0.0005	0.083	0.326	+0.0073	3/3	0.784	128
	Method A	0.615±0.005	(0.611, 0.613, 0.620)	0.0135±0.0001	0.091	0.330	+0.0068	3/3	0.782	128
	output-KL	0.619±0.037	(0.577, 0.638, 0.643)	0.0136±0.0008	0.074	0.336	+0.0072	3/3	0.779	128
JumpReLU	base	0.665±0.011	(0.653, 0.665, 0.676)	0.0146±0.0002	0.111	0.264	+0.0074	3/3	0.811	167
	OWT-only	0.669±0.029	(0.648, 0.658, 0.702)	0.0147±0.0006	0.111	0.255	+0.0074	3/3	0.815	168
	Method A	0.668±0.025	(0.639, 0.685, 0.681)	0.0147±0.0006	0.101	0.265	+0.0075	3/3	0.813	168
	output-KL	0.687±0.014	(0.673, 0.687, 0.701)	0.0151±0.0003	0.059	0.285	+0.0080	3/3	0.819	181

the reconstruction beats matched-norm and decoder-shuffle vectors, and the top features beat 200 matched random feature sets.

Results (Table 13, Table 14, Figure 7).

- **Method A** leaves causal preservation unchanged for every architecture: Method A − base is +0.000 (ReLU), −0.010 (TopK) and +0.004 (JumpReLU), all intervals including zero. E3-H1 is NO IMPROVEMENT in 3/3, and the OWT-only control is also unchanged.
- **Output-KL** raises preservation for ReLU by +0.018 [+0.011, +0.025] over Method A, on every seed and against base and OWT-only too. For JumpReLU the rise is +0.022 over the base (consistent) and +0.018 over Method A (not consistent). Both cells are NOT CREDITABLE: their OWT L0 rose by 5.8 % and 8.1 %, and at the site they use 10 and 13.5 more latents per token.
- **TopK** is the one architecture whose sparsity is fixed and the only creditable output-KL cell. It shows no change (+0.005, interval −0.078 to +0.088; NO IMPROVEMENT).
- **Specificity.** The persona-specific restoration $T_{\text{rec}} - S_{\text{rec}}$ moves by at most 0.0006 nats in any cell, so no fine-tune gains persona-specific causal preservation.
- **Size.** The reconstruction still carries only 0.61–0.69 of the effect after every fine-tune. No cell regresses.

Reading. Neither method creditably improves causal faithfulness at this site. The only gains appear together with extra active latents, and they vanish for the architecture whose L0 cannot move. This matches the representation result for RepE-LM (§7.1) and the behavioural result above: output-KL improves output agreement, but its causal benefit cannot be separated from a sparsity increase.

Table 14: E3 paired comparisons of $T_{\text{rec}}/T_{\text{full}}$ (↑; VALIDATION; 3 seeds; 95% t-interval, df 2) and the pre-registered classes: E3-H1 (Method A, consistent against both base and OWT-only; class shown on the second row) and E3-H2 (output-KL vs Method A; NOT CREDITABLE a priori for ReLU and JumpReLU, whose IMP-L L0 guard failed). Source: `results/repe/e3_finetuned_causal_preservation/comparisons.csv`; `results/repe/e3_finetuned_causal_preservation/criteria.json`.

SAE	Comparison	Δ preservation	95% CI	per seed	$\Delta(T_{\text{rec}} - S_{\text{rec}})$	pre-registered class
ReLU	method_a - base	+0.0002	[-0.0166, +0.0170]	(-0.007, +0.001, +0.007)	+0.00001	
	method_a - owt_only	+0.0006	[-0.0092, +0.0104]	(-0.004, +0.002, +0.003)	+0.00003	NO IMPROVEMENT
	kl - method_a	+0.0180	[+0.0107, +0.0252]	(+0.021, +0.016, +0.016)	+0.00022	NOT CREDITABLE (guard failed)
	owt_only - base	-0.0004	[-0.0087, +0.0078]	(-0.003, -0.002, +0.003)	-0.00001	
	kl - base	+0.0182	[+0.0075, +0.0288]	(+0.015, +0.017, +0.023)	+0.00023	
TopK	method_a - base	-0.0101	[-0.1005, +0.0803]	(+0.031, -0.037, -0.025)	-0.00017	
	method_a - owt_only	-0.0169	[-0.0714, +0.0375]	(+0.007, -0.037, -0.020)	-0.00045	NO IMPROVEMENT
	kl - method_a	+0.0046	[-0.0783, +0.0876]	(-0.034, +0.025, +0.023)	+0.00040	NO IMPROVEMENT
	owt_only - base	+0.0068	[-0.0323, +0.0460]	(+0.025, +0.000, -0.005)	+0.00029	
	kl - base	-0.0055	[-0.0196, +0.0086]	(-0.003, -0.012, -0.002)	+0.00024	
JumpReLU	method_a - base	+0.0040	[-0.0388, +0.0468]	(-0.014, +0.020, +0.005)	+0.00012	
	method_a - owt_only	-0.0008	[-0.0615, +0.0598]	(-0.008, +0.026, -0.021)	+0.00011	NO IMPROVEMENT
	kl - method_a	+0.0184	[-0.0204, +0.0572]	(+0.033, +0.002, +0.019)	+0.00050	NOT CREDITABLE (guard failed)
	owt_only - base	+0.0048	[-0.0410, +0.0506]	(-0.006, -0.006, +0.026)	+0.00002	
	kl - base	+0.0223	[+0.0156, +0.0291]	(+0.019, +0.023, +0.025)	+0.00062	

11 Statistical Analysis

- **Pre-registration.** Every experiment from M0 onwards wrote its hypotheses, criteria and script hashes to `preregistration.json` before running; scripts refuse to overwrite outputs.
- **Within-layer inference (M0).** Fixed-effects regressions with Freedman–Lane permutation within layer (5000 permutations), bootstrap CIs (2000) and Holm correction over four primary pairs.
- **Information (M2).** Statement-grouped bootstrap (paired over the same rows), one-sided tests against noise and PCA references, Benjamini–Hochberg correction across SAEs per task; a 0.02 AUROC tolerance defines "preserved".
- **Causal (M3/M4).** Holm-corrected tests over 65 cells; 200 matched-norm random vectors per cell; synonym and neutral controls; for features, 200 random feature sets matched on activation frequency and $|\Delta z|$.
- **Fine-tuning comparisons (IMP-L, RLM-RP, E3).** Paired by seed within layer and architecture; mean, sd, per-seed values and a 95 % t-interval with 2 degrees of freedom. The pre-registered "consistent" rule (all three seeds and $|\text{mean}| > 2$ sd) is not equivalent to that interval: a cell can meet the rule while its interval touches zero. Per-SAE uncertainty in RLM-RP and E3 comes from a statement-grouped bootstrap (1000 and 2000 resamples). E3's M4 control tests use Benjamini–Hochberg correction over its 36 SAEs. No correction is applied across the 9–12 cells of a fine-tuning criterion.
- **Seeds.** Three training seeds share the same training data, so seed spread understates uncertainty; comparisons that share seeds are paired. The fine-tuning criterion "consistent" requires the improvement on all three seeds and a mean paired difference larger than twice its standard deviation.
- **Splits.** Fitting and selection use training data; validation results are reported as VALIDATION. Only the reconstruction table is TEST.

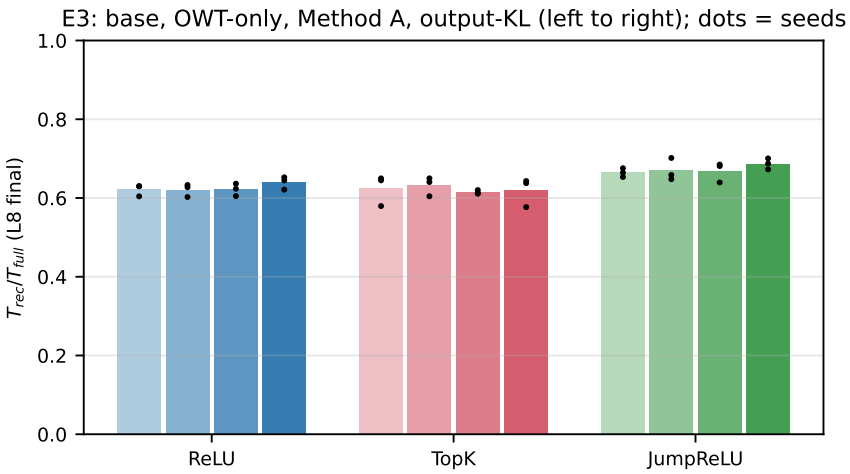


Figure 7: E3 (VALIDATION): share of the persona-instruction effect carried by the SAE reconstruction at the layer-8 final token for base, OWT-only, Method A and output-KL SAEs (bars: mean of 3 seeds; dots: seeds). Output-KL ReLU and JumpReLU violate the pre-registered L0 guard.

PENDING: M5: one pre-registered held-out pass (RepE-LM test, OWT test; GoT test labelled previously exposed), hierarchical bootstrap over statements and seeds, Holm within confirmatory families.

12 Discussion

Reconstruction is a proxy for behaviour, not for information. Within a layer, an SAE with lower reconstruction error damages the model's predictions less, but it does not keep more truth information. Reconstruction metrics average over all directions weighted by their variance; the information that distinguishes true from false statements, or honest from dishonest instructions, lies in small differences between nearly identical activations.

Why the causal effect leaks. M4 makes this concrete. An SAE can reconstruct the honest and the dishonest activation well individually and still reconstruct their difference poorly, because the two reconstruction errors do not cancel. At mid and late sites about a third of the effect is outside the reconstructed difference, and for TopK and JumpReLU part of it exists only when reconstruction and error act together through GPT-2's nonlinear downstream computation.

What fine-tuning changes, and what it does not. The fine-tuning experiments separate three things that are often reported as one. (i) *Task reconstruction* improves reliably: reconstruction error on truth statements falls by 61–85 % in every cell. (ii) *Frozen-probe preservation* improves in direction everywhere and meets the pre-registered criterion in 4 of 9 cells, but a re-trained reader recovers no more truth information from the fine-tuned reconstruction than from the original one (no cell changes consistently). Because M2 showed that frozen-probe gaps are dominated by the probe's sensitivity to off-distribution perturbations, the most economical reading is that reconstruction fine-tuning moves the reconstruction back towards the region where the original probe's decision boundary was fitted, rather than restoring information the SAE had discarded. (iii) *Behavioural preservation* improves with an output-KL term in all 9 cells, of which 5 are creditable (the other 4 fail the sparsity guard); it does not transfer to truth preservation, which is unchanged or slightly

worse. (iv) *Persona representation and causal preservation* are untouched by Method A (RLM-RP and E3: no change in any cell), and output-KL moves them only where it also adds active latents. Improving one level of faithfulness therefore does not improve the others, and a claim of "more faithful SAEs" must name the level.

Sparsity is part of the price. For ReLU and JumpReLU at layers 8 and 9 the output-KL gain comes with a 5–8 % rise in L0 that the continued sparsity controller does not correct. Since lower reconstruction error and lower KL are both easier at higher L0, behavioural gains reported without a sparsity guard would overstate the benefit; TopK, whose L0 is fixed, is the only architecture where the gain is creditable at every layer.

The negative result is the main finding. The SAEs studied here reconstruct well by conventional measures, yet they lose task information that PCA at equal error keeps, and at mid and late layers their reconstruction carries only part of a causal effect that GPT-2 demonstrably carries there. Short fine-tuning improves what it optimises (task reconstruction, output agreement) and nothing we could verify beyond that: E3 finds no creditable gain in *causal* preservation at the layer-8 site, and the only gains it finds (ReLU and JumpReLU under output-KL, about +0.02 of the effect) come with more active latents. All of these are VALIDATION findings.

Persona loss is not only a reader artefact. For truth statements, most of the frozen-probe collapse is reproduced by same-norm noise. For persona instructions it is not: the SAE loses 2–3 times more than the noise does, re-trained readers recover almost none of it, and the per-pair honest-minus-dishonest difference keeps a cosine of only 0.35–0.58. A single word changes these activations, and the SAEs reconstruct the two prompts of a pair with errors that do not cancel. This is consistent with M4: what is lost in representation is also what leaks out of the causal effect.

PENDING: Revise after E1, E2 and M5; in particular, whether the conclusions hold for an externally trained SAE (E2) and at task-level matched sparsity (E1).

13 Limitations and Threats to Validity

- **Small, data-limited SAEs.** 3072 latents (4×), trained on 101,600 document-start tokens for about 38 passes, not converged (validation-based checkpoint selection always chose the final step). The public `gpt2-small-res-jb` SAE used 300M tokens and 24,576 latents. Conclusions may not transfer.
- **One operating point.** L0 = 128 and width 3072 only.
- **Surrogate JumpReLU.** Not the published estimator; up to 85 % dead latents at layer 0.
- **Task-level sparsity mismatch.** Matching is on OWT; task prompts activate very different numbers of latents per architecture.
- **Weak behavioural signal.** The persona effect is 0.02–0.04 nats; the persona-meaning gate passed narrowly (0.701).
- **Seeds share data.** Seed spread understates uncertainty.
- **Test exposure.** The GoT test split was scored in earlier frozen-probe evaluations; the original RepE test split was touched in a smoke run and is retired with the confounded task.

- **No held-out confirmation.** Every result except the reconstruction table is a validation finding; the pre-registered held-out evaluation (M5) has not been run. Validation data were also used to compare methods, so validation effect sizes may be optimistic.
- **Fine-tuning at one budget.** 500 steps at learning rate 10^{-5}, the layer-6 setting, was reused at layers 3, 8 and 9 without tuning; other budgets could change the classes.
- **Three seeds and many cells.** The "consistent" rule (all three seeds, > 2 sd) is strict for variance but is applied to 9 cells per criterion without multiplicity correction; 95 % intervals with two degrees of freedom are wide. Five Method A cells move in the right direction on every seed but fail the 2 sd rule, and the two output-KL regressions meet the 2 sd rule although their 95 % intervals just include zero.
- **Frozen-probe criterion.** The pre-registered truth criterion uses a probe fitted on original activations. Re-trained readers disagree with it (§10.1), so a met criterion is evidence that the original probe reads the reconstruction better, not that information was recovered.
- **Sparsity drift under output-KL.** The continued multiplicative controller does not hold ReLU and JumpReLU L0 within ± 5 % under the KL gradient at layers 8 and 9; an L0-matched variant is untested.
- **Weak truth signal at layer 3.** The original truth MLP probe at layer 3 reaches 0.815 AUROC but the logistic probe only 0.60, so layer-3 truth results rest on a weak signal.
- **Causal preservation after fine-tuning at one site.** E3 covers only the layer-8 final token and a small effect ($T_{\text{full}} = 0.022$ nats); TopK's per-SAE intervals (± 0.06–0.09) cannot resolve changes below about 0.03.
- **Persona task limits.** RepE-LM has no test split; the persona probe transfers only weakly to held-out wording even on original activations (AUROC 0.54–0.69); results are for the final token only; task-level L0 is not matched across architectures (the truncation control is post-hoc, E1 pending); 14 validation statements have opposite-label near duplicates in training (sensitivity ≤ 0.02).
- **Historical RepE results not used.** The original persona prompts had a length shortcut; every RepE result in this paper uses RepE-LM, and the earlier length-confounded results are kept for provenance only.
- **Single model.** No SAE results on Gemma-2-2B [6]; Gemma Scope weights were not available on our machine.

14 Conclusion and Future Work

Across all layers of GPT-2 Small, three SAE architectures reconstruct activations well by conventional measures, but reconstruction quality predicts behavioural damage rather than information preservation, frozen-probe "truth loss" is largely perturbation sensitivity, SAEs never beat PCA at equal error on task information, and at mid and late layers the SAE reconstruction carries only about two-thirds of a causal effect located in the original model. Short fine-tuning improves the faithfulness level it optimises (task reconstruction, or output agreement with an output-KL term) but not truth information as seen by a re-trained reader, and the behavioural gains of ReLU and JumpReLU at depth come with a sparsity cost. On the persona task the SAEs lose information

that re-trained readers cannot recover and distort the honest-minus-dishonest difference, and neither fine-tuning method creditably improves persona representation or the causal effect carried at layer 8. All of these are VALIDATION findings. Future work: larger and longer-trained SAEs, the published JumpReLU estimator, task-level sparsity matching, and the same audit on Gemma-2-2B with Gemma Scope SAEs.

PENDING: Final conclusions after M5.

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A Project History and Corrections

- Layers were first trained with mixed optimiser procedures; all 108 SAEs were retrained with one uninterrupted 15k-step protocol (explained variance changed by at most 0.006; rankings unchanged).
- An architecture summary averaged unmatched runs; it was replaced by a $\pm 10\,\%$ L0 matching rule.
- Earlier "SAE steering" never placed an SAE in the forward pass; it was reclassified.
- The truth-aware fine-tuning objective was circular (§10).
- The original RepE persona prompts were length-confounded; RepE-LM replaces them.
- Correlations pooled across layers were replaced by within-layer analyses (M0).
- Frozen-probe failures were re-analysed with noise and PCA references (M2).

Historical results (original RepE probes and steering, the layer-6 causal test with 200 controls, R1 honesty fine-tuning, R2 feature clamps, Gemma steering, Feature 974) are kept in the repository with their status labels and are not used as evidence here.

B Test-split Exposure

OWT test: used once, for the reconstruction table. GoT test: scored for all 108 SAEs with frozen probes and twice for historical layer-6 fine-tunes. RepE-LM test: never built or accessed. M0–M4, IMP-L, RLM-RP and E3 used no test rows.

C Reproducibility

Code, configurations, pre-registrations and result files: the repository `Fall2026group4/fall-2026-group4`, branch `sae-research`. Software: Python 3.14, PyTorch 2.14.1, TransformerLens 3.9.0 [10], transformers 5.21.0, SAELens 6.54.12. Hardware: one NVIDIA A10G (23 GB). Tables and figures in this paper are generated by `scripts/shared/build_paper_artifacts.py` from the committed result files. SAE checkpoints (about 9 GB) are listed with SHA-256 hashes in `docs/checkpoint_registry.csv`.

Commands for the experiments added on 2026-11-10 (run from the repository root; each refuses to overwrite outputs and checks the hashes in its `preregistration.json`):

- E3: `python scripts/repe/25_e3_finetuned_causal_preservation.py -run` (about 30 min on the A10G);
- RLM-RP: `python scripts/repe/26_repe_lm_representation_preservation.py -run` (about 8 min);
- tables and figures: `python scripts/shared/build_paper_artifacts.py`.

PENDING: Command list per experiment; pipeline diagram (Review #8 F8: `.drawio` source and `.svg`).

C.1 IMP-L paired differences by layer

Mean paired difference over 3 seeds, with the 95 % t-interval beneath (VALIDATION). Generated from `results/geometry_of_truth/multilayer_finetune/comparisons.csv`; per-seed values are in the same file.

Table 15: IMP-L at layer 3 (VALIDATION): paired differences over 3 seeds, mean [95% t-interval]; relative error and L0 as mean percentage change. Lower is better except OWT EV. Source: `results/geometry_of_truth/multilayer_finetune/comparisons.csv`; `results/geometry_of_truth/multilayer_finetune/summary.csv`.

Comparison	Truth gap (B-MLP)	GoT rel. error	OWT EV	OWT ΔCE	OWT KL	GoT stmt. KL	OWT L0
ReLU: base truth gap 0.293, OWT EV 0.904, ΔCE 0.078, OWT L0 134.0							
owt_only - base	-0.001 [-0.006, +0.004]	-1.5%	+0.001 [+0.001, +0.002]	-0.002 [-0.005, +0.002]	-0.002 [-0.003, -0.000]	-0.002 [-0.006, +0.002]	-0.2%
method_a - base	-0.018 [-0.041, +0.005]	-62.0%	+0.001 [+0.000, +0.001]	-0.001 [-0.005, +0.003]	-0.001 [-0.002, +0.000]	-0.001 [-0.007, +0.004]	-0.2%
method_a - owt_only	-0.017 [-0.040, +0.007]	-61.4%	-0.001 [-0.001, -0.001]	+0.001 [-0.000, +0.002]	+0.001 [+0.000, +0.001]	+0.000 [-0.003, +0.004]	+0.0%
kl - base	-0.006 [-0.021, +0.009]	-50.0%	+0.001 [+0.001, +0.001]	-0.009 [-0.013, -0.005]	-0.013 [-0.015, -0.011]	-0.052 [-0.058, -0.046]	+3.5%
kl - method_a	+0.012 [-0.000, +0.024]	+31.5%	+0.000 [+0.000, +0.001]	-0.008 [-0.012, -0.004]	-0.012 [-0.014, -0.010]	-0.050 [-0.051, -0.049]	+3.7%
TopK: base truth gap 0.227, OWT EV 0.904, ΔCE 0.057, OWT L0 128.0							
owt_only - base	+0.000 [-0.008, +0.009]	-3.8%	+0.002 [+0.002, +0.002]	-0.001 [-0.003, +0.001]	-0.001 [-0.002, -0.000]	-0.000 [-0.001, +0.001]	+0.0%
method_a - base	-0.098 [-0.107, -0.089]	-74.0%	+0.001 [+0.001, +0.001]	-0.001 [-0.002, +0.000]	-0.001 [-0.002, -0.000]	-0.000 [-0.001, +0.001]	+0.0%
method_a - owt_only	-0.098 [-0.116, -0.081]	-73.0%	-0.000 [-0.001, -0.000]	+0.000 [-0.001, +0.001]	+0.000 [-0.000, +0.001]	+0.000 [-0.002, +0.002]	+0.0%
kl - base	-0.072 [-0.104, -0.039]	-62.0%	-0.001 [-0.001, -0.000]	-0.003 [-0.005, -0.002]	-0.006 [-0.007, -0.005]	-0.036 [-0.043, -0.030]	+0.0%
kl - method_a	+0.027 [-0.001, +0.054]	+46.2%	-0.002 [-0.002, -0.002]	-0.002 [-0.003, -0.002]	-0.005 [-0.006, -0.005]	-0.036 [-0.043, -0.030]	+0.0%
JumpReLU: base truth gap 0.294, OWT EV 0.914, ΔCE 0.040, OWT L0 131.2							
owt_only - base	+0.000 [-0.014, +0.015]	-1.4%	+0.002 [+0.001, +0.002]	-0.001 [-0.005, +0.002]	-0.002 [-0.003, -0.001]	-0.001 [-0.005, +0.002]	+0.0%
method_a - base	-0.012 [-0.043, +0.019]	-61.1%	+0.001 [+0.001, +0.001]	-0.000 [-0.003, +0.002]	-0.001 [-0.001, -0.001]	-0.001 [-0.002, +0.000]	+0.1%
method_a - owt_only	-0.013 [-0.031, +0.005]	-60.5%	-0.001 [-0.001, -0.001]	+0.001 [-0.000, +0.002]	+0.001 [-0.001, +0.002]	+0.000 [-0.002, +0.002]	+0.1%
kl - base	+0.003 [-0.013, +0.019]	-48.8%	+0.002 [+0.002, +0.002]	-0.006 [-0.010, -0.002]	-0.010 [-0.011, -0.008]	-0.038 [-0.050, -0.026]	+4.6%
kl - method_a	+0.016 [-0.017, +0.048]	+31.5%	+0.001 [+0.001, +0.001]	-0.005 [-0.007, -0.003]	-0.009 [-0.010, -0.007]	-0.037 [-0.048, -0.026]	+4.5%

Table 16: IMP-L at layer 8 (VALIDATION): paired differences over 3 seeds, mean [95% t-interval]; relative error and L0 as mean percentage change. Lower is better except OWT EV. Source: `results/geometry_of_truth/multilayer_finetune/comparisons.csv`; `results/geometry_of_truth/multilayer_finetune/summary.csv`.

Comparison	Truth gap (B-MLP)	GoT rel. error	OWT EV	OWT ΔCE	OWT KL	GoT stmt. KL	OWT L0
ReLU: base truth gap 0.234, OWT EV 0.800, ΔCE 0.223, OWT L0 134.5							
owt_only - base	-0.007 [-0.016, +0.002]	-2.2%	+0.001 [+0.001, +0.002]	-0.002 [-0.002, -0.001]	-0.002 [-0.003, -0.002]	-0.001 [-0.003, +0.000]	+0.1%
method_a - base	-0.054 [-0.095, -0.013]	-73.5%	+0.001 [-0.000, +0.001]	-0.001 [-0.002, -0.000]	-0.001 [-0.002, -0.000]	-0.003 [-0.006, +0.000]	+0.1%
method_a - owt_only	-0.047 [-0.086, -0.008]	-72.9%	-0.001 [-0.001, -0.000]	+0.001 [+0.001, +0.002]	+0.001 [+0.001, +0.002]	-0.002 [-0.004, +0.001]	-0.0%
kl - base	-0.054 [-0.092, -0.016]	-61.9%	+0.003 [+0.002, +0.003]	-0.039 [-0.041, -0.036]	-0.043 [-0.045, -0.042]	-0.154 [-0.238, -0.071]	+5.8%
kl - method_a	-0.000 [-0.037, +0.037]	+43.9%	+0.002 [+0.001, +0.003]	-0.038 [-0.040, -0.036]	-0.042 [-0.044, -0.041]	-0.151 [-0.234, -0.069]	+5.7%
TopK: base truth gap 0.157, OWT EV 0.836, ΔCE 0.126, OWT L0 128.0							
owt_only - base	+0.021 [-0.032, +0.074]	-1.3%	+0.002 [+0.002, +0.002]	-0.002 [-0.004, -0.000]	-0.003 [-0.004, -0.001]	-0.002 [-0.008, +0.004]	+0.0%
method_a - base	-0.074 [-0.202, +0.054]	-85.4%	+0.001 [+0.001, +0.001]	-0.003 [-0.004, -0.001]	-0.002 [-0.004, -0.001]	-0.001 [-0.004, +0.003]	+0.0%
method_a - owt_only	-0.094 [-0.229, +0.040]	-85.2%	-0.001 [-0.001, -0.001]	-0.001 [-0.002, +0.000]	+0.000 [+0.000, +0.001]	+0.002 [-0.005, +0.008]	+0.0%
kl - base	-0.068 [-0.198, +0.062]	-79.6%	-0.002 [-0.002, -0.001]	-0.017 [-0.021, -0.012]	-0.017 [-0.018, -0.016]	-0.079 [-0.103, -0.055]	+0.0%
kl - method_a	+0.006 [-0.065, +0.076]	+40.2%	-0.002 [-0.003, -0.002]	-0.014 [-0.018, -0.010]	-0.015 [-0.016, -0.014]	-0.078 [-0.098, -0.058]	+0.0%
JumpReLU: base truth gap 0.260, OWT EV 0.840, ΔCE 0.117, OWT L0 131.8							
owt_only - base	-0.000 [-0.025, +0.025]	-0.7%	+0.002 [+0.002, +0.003]	-0.003 [-0.005, -0.001]	-0.003 [-0.006, -0.001]	-0.004 [-0.009, +0.002]	+0.4%
method_a - base	-0.060 [-0.247, +0.127]	-79.6%	+0.001 [+0.000, +0.001]	-0.002 [-0.007, +0.002]	-0.002 [-0.004, +0.000]	+0.001 [-0.005, +0.007]	+0.3%
method_a - owt_only	-0.060 [-0.238, +0.119]	-79.4%	-0.002 [-0.002, -0.001]	+0.001 [-0.002, +0.004]	+0.001 [+0.001, +0.002]	+0.005 [-0.005, +0.016]	-0.1%
kl - base	-0.048 [-0.164, +0.067]	-69.4%	+0.005 [+0.004, +0.005]	-0.024 [-0.028, -0.020]	-0.024 [-0.027, -0.021]	-0.102 [-0.114, -0.089]	+8.1%
kl - method_a	+0.011 [-0.065, +0.087]	+49.7%	+0.004 [+0.004, +0.004]	-0.021 [-0.022, -0.021]	-0.022 [-0.024, -0.020]	-0.103 [-0.121, -0.085]	+7.8%

Table 17: IMP-L at layer 9 (VALIDATION): paired differences over 3 seeds, mean [95% t-interval]; relative error and L0 as mean percentage change. Lower is better except OWT EV. Source: `results/geometry_of_truth/multilayer_finetune/comparisons.csv`; `results/geometry_of_truth/multilayer_finetune/summary.csv`.

Comparison	Truth gap (B-MLP)	GoT rel. error	OWT EV	OWT ΔCE	OWT KL	GoT stmt. KL	OWT L0
ReLU: base truth gap 0.214, OWT EV 0.784, ΔCE 0.250, OWT L0 135.0							
owt_only - base	-0.004 [-0.004, -0.003]	-0.7%	+0.001 [-0.000, +0.003]	-0.002 [-0.003, -0.002]	-0.002 [-0.004, -0.001]	+0.001 [-0.009, +0.012]	+0.2%
method_a - base	-0.047 [-0.065, -0.030]	-72.6%	+0.001 [-0.001, +0.002]	-0.002 [-0.003, -0.001]	-0.001 [-0.003, +0.000]	-0.005 [-0.022, +0.011]	+0.1%
method_a - owt_only	-0.044 [-0.062, -0.026]	-72.4%	-0.001 [-0.001, -0.001]	+0.000 [-0.001, +0.001]	+0.001 [+0.000, +0.001]	-0.007 [-0.014, +0.001]	-0.1%
kl - base	-0.015 [-0.069, +0.040]	-59.2%	+0.002 [+0.000, +0.004]	-0.042 [-0.044, -0.041]	-0.046 [-0.048, -0.044]	-0.208 [-0.230, -0.185]	+5.3%
kl - method_a	+0.033 [-0.006, +0.071]	+49.1%	+0.002 [+0.001, +0.002]	-0.041 [-0.042, -0.039]	-0.044 [-0.045, -0.043]	-0.202 [-0.241, -0.164]	+5.2%
TopK: base truth gap 0.160, OWT EV 0.826, ΔCE 0.150, OWT L0 128.0							
owt_only - base	-0.002 [-0.053, +0.050]	-1.6%	+0.002 [+0.002, +0.002]	-0.002 [-0.003, -0.000]	-0.003 [-0.003, -0.002]	-0.002 [-0.009, +0.004]	+0.0%
method_a - base	-0.033 [-0.061, -0.005]	-85.4%	+0.001 [+0.001, +0.001]	-0.002 [-0.003, -0.001]	-0.002 [-0.003, -0.001]	-0.003 [-0.008, +0.002]	+0.0%
method_a - owt_only	-0.031 [-0.103, +0.041]	-85.2%	-0.001 [-0.001, -0.001]	-0.000 [-0.001, -0.000]	+0.000 [-0.000, +0.001]	-0.000 [-0.002, +0.001]	+0.0%
kl - base	-0.023 [-0.028, -0.018]	-80.2%	-0.002 [-0.002, -0.001]	-0.015 [-0.017, -0.013]	-0.018 [-0.018, -0.017]	-0.084 [-0.119, -0.049]	+0.0%
kl - method_a	+0.010 [-0.024, +0.043]	+35.3%	-0.002 [-0.002, -0.002]	-0.013 [-0.016, -0.010]	-0.015 [-0.016, -0.014]	-0.081 [-0.112, -0.050]	+0.0%
JumpReLU: base truth gap 0.258, OWT EV 0.829, ΔCE 0.140, OWT L0 131.9							
owt_only - base	+0.004 [-0.038, +0.047]	-1.1%	+0.002 [+0.002, +0.003]	-0.004 [-0.008, +0.001]	-0.003 [-0.005, -0.002]	-0.003 [-0.011, +0.004]	+0.4%
method_a - base	-0.106 [-0.162, -0.050]	-78.7%	+0.001 [-0.000, +0.001]	-0.002 [-0.008, +0.004]	-0.002 [-0.003, -0.000]	-0.003 [-0.007, +0.001]	+0.3%
method_a - owt_only	-0.110 [-0.169, -0.051]	-78.5%	-0.002 [-0.002, -0.002]	+0.002 [-0.000, +0.004]	+0.002 [+0.001, +0.002]	+0.000 [-0.006, +0.006]	-0.1%
kl - base	-0.084 [-0.095, -0.073]	-70.4%	+0.005 [+0.004, +0.005]	-0.023 [-0.028, -0.017]	-0.025 [-0.027, -0.024]	-0.115 [-0.149, -0.080]	+8.1%
kl - method_a	+0.022 [-0.024, +0.068]	+39.3%	+0.004 [+0.004, +0.005]	-0.021 [-0.022, -0.020]	-0.024 [-0.026, -0.022]	-0.112 [-0.148, -0.075]	+7.7%